GRRR: The Geometry of Reshaping, Rotation, and Routing in Decoder LLM post-training

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Abstract

We study how post-training changes the weights of Large Language Models (LLMs) relative to their pretrained weights. Across 12 post-training chains with supervised fine-tuning (SFT) and reinforcement learning (RL), we express each weight update in the pretrained matrix's singular value decomposition (SVD) frame. This decomposition separates the changes of three geometrically distinct components: diagonal values, which reshapes singular values; off-diagonal values, which rotates the coupling between pretrained input and output directions; and null-space values, which routes outside the matrix's original nonzero SVD core. On a math evaluation suite, we find that removing the diagonal component usually preserves most of the gains from post-training. These results suggest that post-training gains are carried primarily by reconfiguring and extending pretrained pathways rather than by substantially changing singular values of pre-trained models.

1 Introduction

Large-scale pretraining establishes much of an LLM's general capability before post-training begins, and increasing pre-training scale improves loss and downstream performance (Brown et al., 2020; Kaplan et al., 2020; Wei et al., 2022). These findings suggest that broad knowledge and computational capabilities are already encoded in the pretrained weights. Post-training can significantly change model behavior, such as instruction following, reasoning, and human alignment (Zhou et al., 2023; Ouyang et al., 2022; Wu et al., 2024). Despite these large behavioral changes, prior work suggests that post-training often acts as a relatively small perturbation around the pretrained model weight space rather than a full rewrite (Radiya-Dixit and Wang, 2020; Tanwar et al., 2025; Mukherjee et al., 2025b). Recent works investigate this question through singular value decomposition (SVD) analyses of pretrained weights and their updates (Zhu et al., 2025; He and Cao, 2026). In this paper, we ask a more direct question: where do the functionally useful changes introduced by post-training live geometrically in the pre-trained weights space?

Existing spectral analyses of SVD reveal that post-training does more than simply change singular values. Zhu et al. (2025) study Reinforcement Learning with Verifiable Rewards (RLVR) training dynamics and find that its weight matrices are steered away from the principal directions of the pretrained weights. In addition, He and Cao (2026) compare final checkpoints with pre-trained models and find near uniform singular value scaling together with coordinated changes of singular vectors. It naturally raises a follow-up question: in the final weight difference, which parts are functionally necessary for the behavior of post-trained model? Understanding where useful weight changes occur can guide the design of more interpretable and efficient post-training methods that target the components responsible for performance gains. To answer this question, we project the post-trained weight updates onto the pretrained model's SVD coordinates, each weight update projects into SVD's three components: pretrained singular values basis, which are diagonal spectral scaling (**reshaping**); mixture of U and V rotation basis, which are off-diagonal mixing within the pretrained subspace (**rotation**); and those outside of SVD basis, which are null space

route outside SVD basis (**routing**). We call this framework the **G**eometry of **r**eshaping, **r**otation, and **r**outing (**GRRR**).

Our main result shows that post-training gains do not rely on changing pretrained singular values in reasoning setting. Even when aggressive post-training puts more mass in the diagonal channel, removing that channel usually does not hurt downstream performance. GRRR also makes explicit the importance of off-diagonal (rotation) and null space (routing) updates – a substantial part of the update can lie outside the pretrained SVD frame. The useful signal sits mainly in rotation and routing.

We establish this result through causal reconstruction of the partial final weight update for a tested model. For each post-training transition, we selectively add the reshaping, rotation, and routing components to the pretrained checkpoint and evaluate the resulting model. On the 25 transitions where the full model improves over the base model by at least 1 point on the five-benchmark average, keeping only rotation and routing stays within 4.3 points of the full post-trained model in 24 cases. In our setting, the final weight difference changes behavior mainly by remixing pretrained features and routing beyond the pretrained frame, not by scaling singular values.

Contributions. We make three contributions. First, we introduce GRRR, a decomposition of post-training weight differences into diagonal reshaping, rotation, and routing. GRRR extends SVD-based weight analysis from descriptive geometry to downstream performance: by adding each component back to the pretrained model separately, we can test which components actually carry the performance gain. Second, we use that causal view to show that removing the diagonal from the final weight difference usually does not hurt on our shared evaluation suite, with one clear outlier involving heavy distillation. In most of the settings we study, the useful signal sits mainly in rotation and routing. Third, we show that the geometry is structured: incremental RL rotations are nearly orthogonal to the preceding SFT rotations, while weight updates produced by independently trained RL and SFT from the same base show weak alignment. Other than that, we do not see clear algorithm label differences reflected on weights difference.

Study scope. Our empirical scope is specific. All models in the paper are decoder LLMs. The main causal evaluation uses a five benchmark average dominated by math reasoning. We therefore limit our strongest claims to decoder LLM post-training in these settings, rather than to all post-training regime.

2 Related Work

Spectral Analysis of Neural Network Weights. SVD has long been used to study neural network training. Martin and Mahoney (2018) use random matrix theory to characterize how weight spectra evolve across training. Yunis et al. (2024) extend this view to modern architectures and track both singular values and singular vectors. For post-training, He and Cao (2026) analyze full weight matrices and argue that post-training often looks like near uniform singular value scaling, akin to a temperature shift, together with coordinated rotations of singular vectors. Zhu et al. (2025) study RLVR training dynamics through leading singular components and argue that RL is steered away from principal or diagonal directions while SFT more often targets them. GRRR is complementary to both. We decompose the final update ΔW in the pretrained basis rather than comparing full weights in their own coordinates. That keeps one shared frame across methods, supports add/remove causal ablations on the final checkpoint, and separates a null term outside the pretrained frame that full matrix SVD comparisons leave implicit.

SVD-Based Parameter Efficient Fine Tuning. A parallel line of work uses the pretrained SVD basis to design efficient adapters. LoRA (Hu et al., 2022) constrains updates to low rank. PiSSA, MiLoRA, and DoRA all use SVD structure in different ways (Meng et al., 2024; Wang et al., 2025; Liu et al., 2024). SVFT (Lingam et al., 2024) is closest to our setup. It parameterizes the weight update in the pretrained SVD basis and finds that training the full singular value matrix can outperform other LoRA configurations. Our goal is different.

We do not use the decomposition for efficiency. We use it to ask which geometric part of the update carries the behavioral change.

post-training Weight Changes. Several works study the structure of post-training weight changes. Ilharco et al. (2023) define task vectors and find low alignment across tasks. Shuttleworth et al. (2025) show that LoRA can introduce new directions outside the pretrained spectrum. Si et al. (2025) find that fine-tuning changes vector orientation more than singular values. On the SFT versus RL question, Mukherjee et al. (2025a) show that RL updates only 5–30% of parameters across seven algorithms, while Jin et al. (2025) argue that singular vector direction matters more than singular values for out of distribution (OOD) performance. GRRR ties these observations together. The diagonal matters less than it appears to, and the split between the off-diagonal term and the null space term helps explain why RL can look sparse while SFT looks larger.

3 The GRRR Framework

Existing spectral analyses decompose each trained model in its own coordinates which makes relative differences very difficult to compare. To make post-trained models comparable, we use the pretrained model as the shared coordinate system. This makes different post-trained models starting from the same pretrained model comparable with a same pretrained model coordinate. This gives one basis for every update and splits the change into three parts: diagonal reshaping, off-diagonal remixing, and null space routing. GRRR is a shared geometric decomposition that makes these updates comparable and supports direct causal tests.

The SVD Decomposition of Weight Changes We can decompose the pre-trained weight matrix $\mathbf{W} \in \mathbb{R}^{m\times n}$ with a thin SVD:

$$\mathbf{W} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top, \quad \mathbf{U} \in \mathbb{R}^{m\times r},\ \mathbf{\Sigma} = \mathrm{diag}(\sigma_1, \ldots, \sigma_r),\ \mathbf{V} \in \mathbb{R}^{n\times r}, \tag{1}$$

where $r = \min(m, n)$ and $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r \geq 0$. We view each singular triple (u_i, σ_i, v_i) as one route from an input pattern to an output pattern. The singular value σ_i says how strong that route is. Changing the diagonal part in $\mathbf{\Sigma}$ changes route strength, but changing the off-diagonal of $\mathbf{\Sigma}$ to none-zero means it changes how inputs are matched to outputs of $\mathbf{U}$ or $\mathbf{V}$. We use the *thin* SVD of the pretrained matrix because it gives a shared orthonormal frame with $r = \min(m, n)$ for every update built from the same base, and we can nicely separate out the basis within the pretrained coordinates and the null residual outside of SVD basis.

The Perturbation Matrix Given a pretrained base weight $\mathbf{W}_0 = \mathbf{U}_0\mathbf{\Sigma}_0\mathbf{V}_0^\top$ and a post-trained weight $\mathbf{W}_1$, we can express the update as $\Delta\mathbf{W} = \mathbf{W}_1 - \mathbf{W}_0$. We project the update into the pretrained SVD coordinate as **perturbation matrix** because pretrained model is shared across many different post-trained models:

$$\mathbf{P} = \mathbf{U}_0^\top(\Delta\mathbf{W})\mathbf{V}_0 \ \in \ \mathbb{R}^{r\times r}. \tag{2}$$

The matrix $\mathbf{P}$ is the update written in the base model's SVD coordinates. Equivalently,

$$\mathbf{U}_0^\top\mathbf{W}_1\mathbf{V}_0 = \mathbf{\Sigma}_0 + \mathbf{P},$$

and we can see very clearly $\mathbf{W}_1 = \mathbf{U}_0(\mathbf{\Sigma}_0 + \mathbf{P})\mathbf{V}_0^\top$. So $\mathbf{P}$ tells us how the post-training weight departs from the base model inside the pretrained frame. Then we build three components:

$$\Delta\mathbf{W}_{\text{diag}} = \mathbf{U}_0\,\mathrm{diag}(\mathbf{P})\mathbf{V}_0^\top, \tag{3}$$

$$\Delta\mathbf{W}_{\text{off}} = \mathbf{U}_0\big(\mathbf{P} - \mathrm{diag}(\mathbf{P})\big)\mathbf{V}_0^\top, \tag{4}$$

$$\Delta\mathbf{W}_{\text{null}} = \Delta\mathbf{W} - \mathbf{U}_0\mathbf{P}\mathbf{V}_0^\top. \tag{5}$$

These correspond to three simple kinds of change: (1) $\Delta\mathbf{W}_{\text{diag}}$: **diagonal spectral reshaping**. Changes singular value magnitudes within the pretrained subspace. (2) $\Delta\mathbf{W}_{\text{off}}$: **off-diagonal rotation**. Rotates pretrained singular directions within the same subspace (off-diagonal structure in $\mathbf{P}$). (3) $\Delta\mathbf{W}_{\text{null}}$: **null space routing**. The residual outside the pretrained SVD frame. While $\mathbf{U}_0\mathbf{P}\mathbf{V}_0^\top$ is the orthogonal projection of $\Delta\mathbf{W}$ onto the pretrained subspace, $\Delta\mathbf{W}_{\text{null}}$ captures all components that cannot be expressed in that basis.

For readability, we refer to $\Delta\mathbf{W}_{\text{off}}$ as *rotation* and $\Delta\mathbf{W}_{\text{null}}$ as *routing* throughout.

By construction, these three components are exactly orthogonal:

$$\|\Delta\mathbf{W}\|_F^2 = \left\|\Delta\mathbf{W}_{\text{diag}}\right\|_F^2 + \|\Delta\mathbf{W}_{\text{off}}\|_F^2 + \|\Delta\mathbf{W}_{\text{null}}\|_F^2 . \tag{6}$$

This three way split matters because the pretrained SVD gives a shared reference frame across post-training methods. In that frame, the diagonal term rescales existing routes, the off-diagonal term mixes them, and the null term leaves the in frame part of the layer. We can use the decomposition for causal tests by adding each part back separately and checking which one actually carries the gain. Under the thin SVD definition above, when a weight matrix is square, $\mathbf{U}_0$ and $\mathbf{V}_0$ span the ambient spaces and the null term vanishes. In many transformer weights, however, the projections are rectangular and null part is significant.

The Interpretation of Components Each projection matrix $\mathbf{W}_0 \in \mathbb{R}^{m\times n}$ maps hidden states to outputs via $\mathbf{W}_0 h = \mathbf{U}_0\mathbf{\Sigma}_0(\mathbf{V}_0^\top h)$. From a different view, we can treat the right singular vectors as specifying which hidden state directions this layer reads from, the left singular vectors as specifying which output directions it writes into, and the singular values as determining the gain of these read-write routes. The diagonal and off-diagonal components both lie in

$$\mathcal{S} = \{\mathbf{U}_0 A \mathbf{V}_0^\top : A \in \mathbb{R}^{r\times r}\},$$

so they only rescale or remix routes within the pretrained singular vector frame. The $\Delta\mathbf{W}_{\text{diag}}$ changes rescale the strength of the singular values and the $\Delta\mathbf{W}_{\text{off}}$ changes the mixture of the read-write routes within the frame. Instead of mapping v_i primarily to u_i, off-diagonal mixes v_i into u_j for $i \neq j$. So the layer reuses its pretrained features but mixes input features with different outputs. The $\Delta\mathbf{W}_{\text{null}}$ component captures updates outside the nonzero pretrained SVD core.

For square projections, $\Delta\mathbf{W}_{\text{null}}$ vanishes identically under the thin SVD definition because $\mathbf{U}_0$ and $\mathbf{V}_0$ span the full ambient spaces. For rectangular matrices, the null term is the part of the update that cannot be written as $\mathbf{U}_0 A \mathbf{V}_0^\top$, and therefore lies outside that pretrained frame. Concretely, $\Delta\mathbf{W}_{\text{null}}$ lets the layer read from hidden state directions outside $\text{col}(\mathbf{V}_0)$ or write into output directions outside $\text{col}(\mathbf{U}_0)$, depending on which side is rank deficient. As an interpretation, it means if the features are in the null space, the weight matrix has zero process of it and adds nothing on top of the residual. Changing in null simply means it allows the layer to read and write features that the pretrained layer ignored, and it allows the post-trained weight matrix to express pretrained matrix could not express.

We want to emphasize this as a local geometric statement about one layer rather than claiming the network as a whole creates new features. Residual paths may already represent or process the local null features in other layers. Our claim is that, relative to the pretrained map of this layer, $\Delta\mathbf{W}_{\text{null}}$ accesses directions outside the layer's original SVD basis. Empirically, post-training usually preserves stable rank and keeps the leading singular value structure highly aligned with the pretrained weight. This suggests that post-training can change where a layer reads and writes without greatly increasing the number of dominant modes. We analyze the effective rank in Appendix D.

3.1 Experimental Setup

Models and transitions. We study 12 training chains spanning 8 architecture families: Qwen 2.5, Qwen 3, Mistral, LLaMA 3, OLMo 2, OLMo 3, DeepSeek, and MiMo. Model

sizes range from 1.5B to 14B parameters (Qwen et al., 2025; Yang et al., 2025; Jiang et al., 2023; Grattafiori et al., 2024; OLMo et al., 2025; Olmo et al., 2025; Guo et al., 2025; Shao et al., 2024; Xiaomi et al., 2025). The chain identifiers run from C1 to C12. The main inventory contains 29 incremental transitions. Some appendices also include cumulative base to checkpoint transitions, which brings the count to 42 transition instances. The post-training methods cover SFT, DPO, PPO, GRPO, GPPO, RAFT, PRIME, and OlmoRL (Rafailov et al., 2024; Schulman et al., 2017; Shao et al., 2024; Su et al., 2026; Zhang et al., 2025a; Cui et al., 2025; Olmo et al., 2025). Domains range from math reasoning to general instruction following and agent tasks. The full inventory appears in Appendix B, Tables 1–2.

Decomposition scope. For each transition we decompose all seven projection matrices (q, k, v, o, gate, up, down) in every transformer layer. Each matrix uses its own thin SVD of the pretrained weight, so the diagonal, off-diagonal, and null components are defined matrix by matrix. In the causal experiments, each component is added back independently to its corresponding base matrix, and the resulting model applies that replacement across all decomposed matrices. Unlike prior work that focuses on top-K structure, we use the full decomposition of each matrix.

Causal evaluation. For the causal decomposition in Section 4.2, we build four ablated variants per transition: diagonal only, off only, null only, and an OffNull variant that keeps the off and null terms together. We compare them against the full update on five math benchmarks: MATH500, AIME2024, AMC2023, Minerva-Math, and OlympiadBench, all at temperature 0 using Zhang et al. (2025b). We use the same evaluation harness for every ablation and report the unweighted average across the five benchmarks. For general instruction checkpoints, this should be read as recovery on a shared math-heavy probe suite rather than on the checkpoint's original training objective. Appendix E reports 28 transition level decompositions (11 SFT, 17 RL) across 11 chains. We exclude C4 web research agent since it is unrelated to math benchmarks.

We report all transitions in the appendix tables and raw plots. When we summarize *recovered improvement*, we use only transitions where the full post-trained model improves over base by at least 1 point on the five benchmark average. This avoids unstable percentages near zero gain and separates genuine recovery of improvement from flat or negative gain cases. Under this rule, the informative set has 25 transitions. C10 SFT, C5 zephyr-DPO, and C7 RLVR remain in the appendix for completeness but are not used as headline evidence about recovered gain.

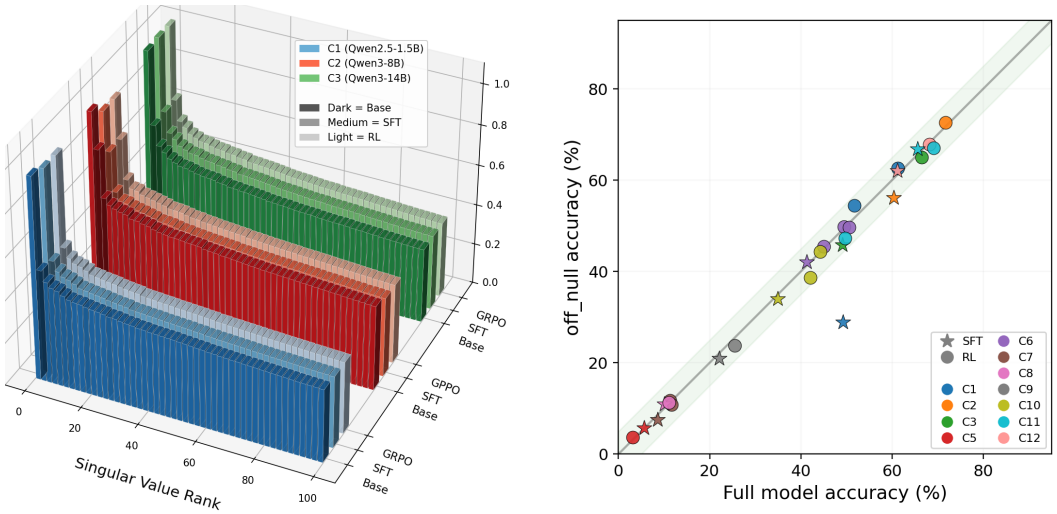
4 The Diagonal Usually Does Not Improve the Performance

We start with the most common spectral question. Does post-training work by reshaping the pretrained spectrum, and does that reshaping matter? In most cases, no. Across models, the singular value profile before and after post-training is almost unchanged, even when behavior changes a lot. Removing the diagonal spectral scaling term also usually leaves performance nearly intact.

4.1 post-training Usually Preserves the Spectrum

Prior work uses spectral structure to explain different aspects of post-training. Zhu et al. (2025) finds that RL training is steered away from principal or diagonal directions, while He and Cao (2026) finds near uniform singular value scaling in post-training with coordinated rotations. Our post-hoc result is complementary to previous studies, and we find the singular value profile before and after post-training is extremely similar, and the leading singular values are almost unchanged (Figure 1a). Even when the update is relatively large (relative weights change values are in Table 3), post-training rarely changes the pretrained spectrum.

This does not mean the diagonal update is always zero. Larger updates can place more energy in the diagonal channel as shown in Appendix F, and SFT often has larger relative updates than RL. We do not see $\Delta\mathbf{W}_{\text{diag}}$ in the model chains we study can cleanly separate SFT from RL, nor does it meaningfully reshape the pretrained spectrum. We find simi-



(a) Top 100 singular values from base, SFT, RL.

(b) Removing the diagonal almost never hurts.

Figure 1: Left: the leading singular values remain visually almost identical before and after post-training. We plot three chains with large spectral changes. Right: Across nearly all tested transitions, removing the diagonal reproduces the full post-trained model with trivial degradation with the exception of C1.

lar results as He and Cao (2026) that large $\Delta\mathbf{W}_{\text{diag}}$ shows a near uniform scale shift of $\mathbf{\Sigma}$. Figure 1a shows the normalized spectra for the chains with the largest changes. We find the profiles are nearly identical. To measure quantitatively, we turn the $\mathbf{\Sigma}$ into vector and measure the cosine similarity between the pretrained and post-trained singular values. We find the spectral cosine exceeds 0.9999 for every transition. The spectrum profile is effectively unchanged. Detailed results are in Appendix C. This suggests that the useful part of post-training may lie less in $\Delta\mathbf{W}_{\text{diag}}$ and more in how computation is rerouted within and beyond the pretrained subspace.

4.2 Removing the Diagonal Usually Does Not Hurt

The spectral analyses above are descriptive as they only show modest difference of singular-value profile, but they do not establish whether the diagonal component is functionally important. We therefore test its causal contribution by removing ΔW_{diag} from the post-trained model and reevaluating performance. Equivalently, it means keeping only $\Delta\mathbf{W}_{\text{off}} + \Delta\mathbf{W}_{\text{null}}$ on top of the base model. This intervention measures how much of the post-training improvement depends on the learned diagonal component. Figure **??** reports the average performance on our math-heavy evaluation suite, while Appendix **??** provides the complete benchmark-level results. Across the 25 positive gain transitions defined in Section 3.1, the model without $\Delta\mathbf{W}_{\text{diag}}$ stays within 4.3 points of the full post-trained model in 24 cases. The single clear exception is C1 distillation SFT, where removing the diagonal change causes a 20.3-point drop.

C1 is also an outlier in the geometry of the update. It distills the 671B DeepSeek-R1 into a 1.5B Qwen2.5-Math-1.5B base (Guo et al., 2025), making it the most aggressive distillation setting in our study. It has the highest diagonal energy among SFT transitions and one of the largest spectral perturbations (Appendix C and Appendix F), and it is the only case where we see removing the $\Delta\mathbf{W}_{\text{diag}}$ causes a large behavioral failure. Inspection of the generated outputs suggests that the $\Delta\mathbf{W}_{\text{diag}}$ in C1 helps control the model's self-reflection behavior. After $\Delta\mathbf{W}_{\text{diag}}$ removal, the model frequently enters extended reflection loops, repeatedly changing direction without producing a final answer. The average number of "Wait" tokens rises from 46.4 to 171.7 per answer and model keeps changes direction without giving a final answer. This suggests that the diagonal change in C1 distillation SFT carries behaviorally important information. We treat it as a heavy distillation outlier rather

than evidence that the diagonal is generally necessary, and other 24 cases do not exhibit a comparable dependence on ΔW_{diag}. Moreover, the subsequent DeepScaleR and Nemotron GRPO transitions initialized from the C1 distilled checkpoint do not show the same diagonal dependence.

Is it the removing size or the diagonal subspace? In most transitions, the $\Delta\mathbf{W}_{\text{diag}}$ contains only a small fraction of the total update energy. Consequently, good performance after $\Delta\mathbf{W}_{\text{diag}}$ removal could simply reflect that little update energy was removed, rather than that the diagonal direction is unusually unimportant. To distinguish component magnitude from component direction, we compare diagonal removal with a same-norm $\left|\Delta W_{\text{diag}}\right|_F$ random-removal control. We test it on two chains, C1 and C2, since they carry substantial diagonal energy, whereas other transitions carry at most 0.011. The results are in Appendix H Table 8.

The two chains are in opposite directions. In C2, removing the $\Delta\mathbf{W}_{\text{diag}}$ costs 4.3 points while removing the same energy in a random direction costs 15.2. It implies that removing the diagonal is cheaper than removing a random direction. C1 is the opposite, as expected, since we show in the paper that the diagonal is important for C1. In the remaining transitions, both diagonal removal and same-norm random removal have small effects within noise ($\pm 3\%$), so we cannot discriminate between them there.

These results distinguish diagonal *change* from diagonal *contribution*. A large diagonal component need not carry a large fraction of the performance gain, as shown by C2, while C1 demonstrates that the diagonal can matter under an exceptional heavy-distillation shift. Overall, however, diagonal reshaping is not the necessary carrier of most post-training gains in the transitions we study: removing it preserves nearly all of the full model's performance in 24 of 25 positive-gain cases. The remaining gains are retained primarily by off-diagonal mixing within the pretrained SVD core and null-space routing outside it, whose relative roles we examine next.

5 Useful Changes Live in the Off-Diagonal and Null Space Terms

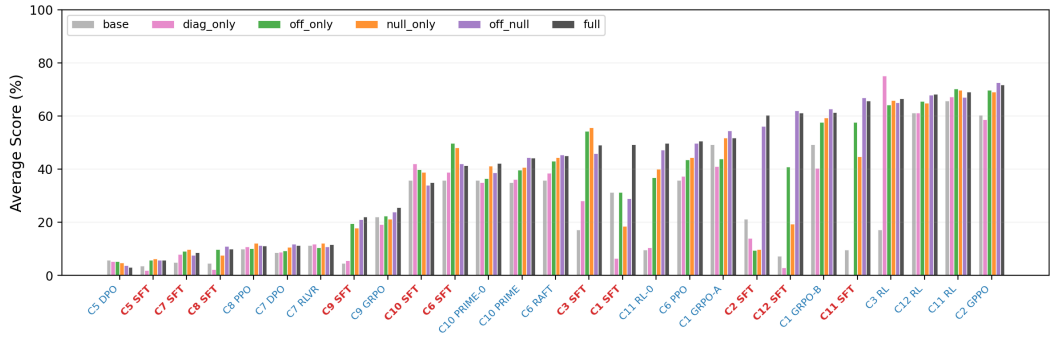
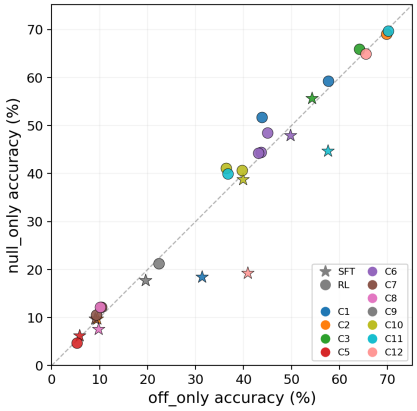


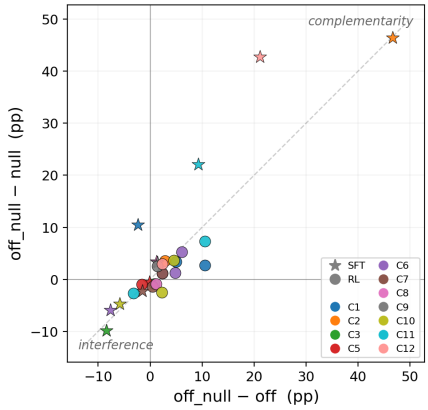
Figure 2: **Ablation scores relative to base and full model.** Each bar is one transition, sorted by full score. The gray bar marks the base model score; the black bar marks the full post-trained model score. Colored markers show the performance of each isolated component: diagonal only (pink), off-diagonal only (green), null space only (orange), and OffNull (purple). Red names correspond to SFT models, and blue names are RL models.

GRRR leaves two candidates of performance gain after the diagonal is removed: off-diagonal updates ($\Delta\mathbf{W}_{\text{off}}$), which remix existing singular directions, and null space updates ($\Delta\mathbf{W}_{\text{null}}$), which route outside the pretrained frame. We ask two concrete questions: how much does each component recover on its own, and what changes when we combine them? We analyze all 28 causal decompositions. We follow Section 3.1, and it leaves 25 informative transitions: 10 SFT and 15 RL. Figure 2 plots the raw five benchmark scores for the full set.

Individual recovery. The OffNull model recovers nearly the full post-training gain in almost every informative case. We do not see a clean split by SFT/RL algorithm differences, and the more consistent pattern is redundancy versus asymmetry. RL transitions often recover well from either component alone, so off and null redundantly store updates. SFT transitions show wider variance and stronger asymmetry between the two components, with the dominant channel varying by chain. At the pooled level, SFT more often has one component clearly ahead of the other, while RL more often places the two components near parity. Figure 3a shows this more directly: RL points cluster near the equal contribution diagonal, while SFT points are more spread out and often lean toward the off-diagonal component for performance gain.



(a) **Off vs null component performance.** Each point is one transition plotted by its off-diagonal performance against its null space performance in raw accuracy.



(b) **Interaction between off-diagonal and null space updates.** Positive values indicate complementarity. Negative values indicate interference.

Figure 3: Left: off vs null accuracy. Right: interaction of off and null

Interaction. We next ask whether keeping both off and null terms adds anything beyond either component alone. Let OffNull denote the model that keeps the off and null terms together. We track the two signed quantities

$$(\text{OffNull} - \text{off}) \quad \text{and} \quad (\text{OffNull} - \text{null}), \tag{7}$$

where positive values indicate complementarity and negative values indicate interference. Values near zero mean that the combination adds little over the stronger single component. Figure 3b shows the result. In RL, most values are small and slightly positive. Off and null carry largely redundant information, so combining them gives a modest but reliable gain. In SFT, the distribution is wider, more asymmetric, and sometimes negative. Some transitions benefit substantially from retaining both components, while in others an individual component can outperform their combination. Once the diagonal is removed, the gain lives in rotation and routing. RL updates more often show redundancy between the two channels, whereas SFT updates exhibit greater asymmetry and stronger interaction. We next ask whether successive stages of post-training reuse the same update directions or move into different parts of the weight space.

6 Sequential SFT and RL Have Nearly Orthogonal Rotations

We do not see the SFT and RL difference geometrically from the section above, but are there other differences? The GRRR framework allows us to compare the SFT and RL under the same base model coordinates. Modern pipelines usually follow a sequence with SFT first, then RL. This makes the geometry of the pipeline interesting to study. Jin et al. (2025) argue that RL softly realigns singular vectors that SFT had rotated. Mukherjee et al. (2025a) find that RL touches only 5–30% of parameters, which suggests limited overlap with SFT. GRRR allows us to ask directly whether the SFT and incremental RL weight updates are aligned, opposed, or nearly orthogonal.

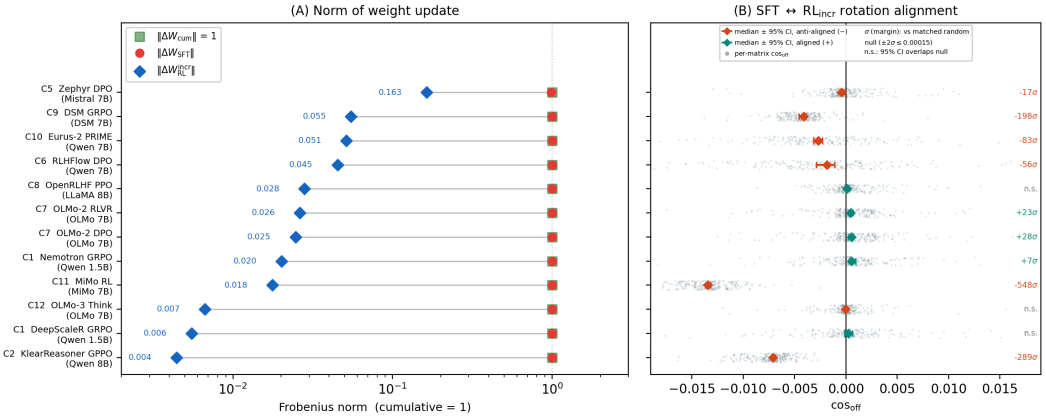


Figure 4: **(A)** Normalized Frobenius norms (log scale) of the SFT, incremental RL, and cumulative updates. SFT dominates in all chains. **(B)** Off-diagonal cosine similarity between SFT and incremental RL rotations. Diamonds show the median over matrices and bars show bootstrap 95% confidence intervals. Significance is measured against a matrix- and chain-matched isotropic null for the same median statistic.

Because both SFT and RL updates are projected into the same pretrained SVD basis, we can compare them directly. We study the standard sequential setting (base $\rightarrow$ SFT $\rightarrow$ RL) and decompose the cumulative update as $\Delta W_{\text{cum}} = \Delta W_{\text{SFT}} + \Delta W_{\text{incr}}$ where $\Delta W_{\text{SFT}} = W_{\text{SFT}} - W_{\text{base}}$ and $\Delta W_{\text{incr}} = W_{\text{RL}} - W_{\text{SFT}}$.

SFT dominates update magnitude. Figure 4A shows the Frobenius norm of each component, normalized so that $\|\Delta W_{\text{cum}}\| = 1$. Across all chains, SFT accounts for nearly the entire update magnitude. The incremental RL step is one to two orders of magnitude smaller (≈ 0.004–$0.17\times$). In raw size, the pipeline is dominated by SFT.

RL Rotations Are Almost Orthogonal to SFT Rotations Magnitude does not determine direction, so we compare the off-diagonal components of the two updates. The rotation are the off-diagonal P-matrix in the pretrained SVD frame. We compute: $\cos_{\text{off}} = \frac{\langle P^{\text{off}}_{\text{SFT}}, P^{\text{off}}_{\text{incr}} \rangle_F}{\|P^{\text{off}}_{\text{SFT}}\|_F \cdot \|P^{\text{off}}_{\text{incr}}\|_F}$ as the similarity measure of the post-trained off-diagonal changes. Figure 4B shows that the median cosine remains small in absolute value. However, near-zero cosine alone is not sufficient evidence because random vectors are already nearly orthogonal in high-dimensional spaces. The key is whether the cosine is smaller than chance. We therefore compare each chain against an isotropic null matched to its own matrix shapes, and to the matrices statistic. We plot the result in the Figure 4B. Because cosine is scale invariant, matching the matrix dimensions and aggregation statistic determines the appropriate null.

Under this baseline, 9 of the 12 sequential transitions differ significantly from chance. 6 are significantly negative, 3 are weakly positive, and 3 are statistically indistinguishable from the null. The negative pattern is also consistent across matrices. The negative values are not explained by high-dimensional noise. Their small absolute magnitude indicates partial reversal rather than strong opposition. This observation is consistent with Jin et al.'s observation that RL can partially reverse rotations induced by SFT. The more informative comparison is with SFT and RL updates trained independently from the same base model. As shown in Appendix G Figure 5 B, independent SFT and RL updates are substantially more aligned when we compare their cosine values, whereas incremental RL following SFT is nearly orthogonal or weakly anti-aligned. More discussions are in Appendix G and comparing sequential and independent trained SFT/RL is more informative.

7 Discussion and Limitations

Our study is limited to decoder LLMs, and our strongest performance claims come from a five-benchmark average dominated by math reasoning. We study fully trained checkpoints and the final weight difference ΔW, not training-time interventions or optimization trajectories. Our causal claims are about post-hoc ablations of the final ΔW. We also study only the linear weight matrices; activations, normalization layers, optimizers, and training dynamics are outside the scope of this paper.

The following are three observations from the weight space. (1) Independent SVDs remain almost unchanged after post-training (Appendix C), effective rank of matrices are stable (Appendix D), and removing the diagonal usually does not hurt performance. Together, these results suggest that post-training usually remains near an approximately isospectral neighborhood of the pretrained model. The main functional change is not a reconstruction of the singular spectrum, but a reorganization of how the pretrained weight space is used, specifically in the off-diagonal and null terms. (2) At the matrix level, the diagonal changes the strength of matched singular vector channels. Off-diagonal terms remix those singular vector channels. Null terms change what the matrix can see and write, and they let a layer to read/write hidden state directions outside its pretrained input subspace. It allows the layer to read/write features previously ignored. (3) The sequential SFT then RL pipeline adds another constraint. We see a contrast between the sequential and the independent SFT/RL. We believe the algorithm label alone is insufficient, and controlled data and compute are more important as the controlled C3 SFT and RL runs are nearly identical in both direction and magnitude in Figure 5. Sequential SFT/RL show a more interesting relationship in weight space. When RL starts after SFT, it tends to move into directions that the preceding update did not use and sometimes partially reverses it, rather than simply amplifying the same rotation.

8 Conclusion

We introduce GRRR, a simple decomposition of post-training weight changes in the pretrained SVD frame into diagonal reshaping, off-diagonal rotation, and null space routing. This decomposition extends SVD-based weight analysis from description to function. GRRR reconstructs models from each component and directly tests which parts of the final weight difference carry the resulting behavior. Across the decoder LLM settings we study, the main result is that the diagonal spectral reshaping is usually not the necessary carrier of post-training gains. Most ordinary post-training gains do not require rebuilding spectral values except heavy distillation. The useful change is retained mainly through two forms of reorganization – singular vectors and null space. Together, these results suggest a distinction between the roles of pretraining and post-training: pretraining constructs the model's weight-space structure, while post-training primarily reorganizes how that structure is used. This view opens several directions. Orthogonal or weakly interfering update subspaces may enable more reliable model merging and weight arithmetic; separating changes in the left and right singular directions may clarify how post-training modifies what a layer reads and writes; and finer analysis across attention, MLP, and other matrix types may reveal where different forms of routing occur. GRRR provides a framework for studying post-training as the organization of pretrained capacity rather than its reconstruction.

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References

Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, Sandhini Agarwal, Ariel Herbert-Voss, Gretchen Krueger, Tom Henighan, Rewon Child, Aditya Ramesh, Daniel Ziegler, Jeffrey Wu, Clemens Winter, Chris Hesse, Mark Chen, Eric Sigler, Mateusz Litwin, Scott Gray, Benjamin Chess, Jack Clark, Christopher Berner, Sam McCandlish, Alec Radford, Ilya Sutskever, and Dario Amodei. Language models are few-shot learners. In H. Larochelle, M. Ranzato, R. Hadsell, M.F. Balcan, and H. Lin, editors, *Advances in Neural Information Processing Systems*, volume 33, pages 1877–1901. Curran Associates, Inc., 2020. URL https://proceedings.neurips.cc/paper_files/paper/2020/file/1457c0d6bfcb4967418bfb8ac142f64a-Paper.pdf.

Shiyi Cao, Dacheng Li, Fangzhou Zhao, Shuo Yuan, Sumanth R. Hegde, Connor Chen, Charlie Ruan, Tyler Griggs, Shu Liu, Eric Tang, Richard Liaw, Philipp Moritz, Matei Zaharia, Joseph E. Gonzalez, and Ion Stoica. Skyrl-agent: Efficient rl training for multi-turn llm agent, 2025. URL https://arxiv.org/abs/2511.16108.

Ganqu Cui, Lifan Yuan, Zefan Wang, Hanbin Wang, Yuchen Zhang, Jiacheng Chen, Wendi Li, Bingxiang He, Yuchen Fan, Tianyu Yu, Qixin Xu, Weize Chen, Jiarui Yuan, Huayu Chen, Kaiyan Zhang, Xingtai Lv, Shuo Wang, Yuan Yao, Xu Han, Hao Peng, Yu Cheng, Zhiyuan Liu, Maosong Sun, Bowen Zhou, and Ning Ding. Process reinforcement through implicit rewards, 2025. URL https://arxiv.org/abs/2502.01456.

Hanze Dong, Wei Xiong, Bo Pang, Haoxiang Wang, Han Zhao, Yingbo Zhou, Nan Jiang, Doyen Sahoo, Caiming Xiong, and Tong Zhang. Rlhf workflow: From reward modeling to online rlhf, 2024. URL https://arxiv.org/abs/2405.07863.

Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Alex Vaughan, Amy Yang, Angela Fan, Anirudh Goyal, Anthony Hartshorn, Aobo Yang, Archi Mitra, Archie Sravankumar, Artem Korenev, Arthur Hinsvark, Arun Rao, Aston Zhang, Aurelien Rodriguez, Austen Gregerson, Ava Spataru, Baptiste Roziere, Bethany Biron, Binh Tang, Bobbie Chern, Charlotte Caucheteux, Chaya Nayak, Chloe Bi, Chris Marra, Chris McConnell, Christian Keller, Christophe Touret, Chunyang Wu, Corinne Wong, Cristian Canton Ferrer, Cyrus Nikolaidis, Damien Allonsius, Daniel Song, Danielle Pintz, Danny Livshits, Danny Wyatt, David Esiobu, Dhruv Choudhary, Dhruv Mahajan, Diego Garcia-Olano, Diego Perino, Dieuwke Hupkes, Egor Lakomkin, Ehab AlBadawy, Elina Lobanova, Emily Dinan, Eric Michael Smith, Filip Radenovic, Francisco Guzmán, Frank Zhang, Gabriel Synnaeve, Gabrielle Lee, Georgia Lewis Anderson, Govind Thattai, Graeme Nail, Gregoire Mialon, Guan Pang, Guillem Cucurell, Hailey Nguyen, Hannah Korevaar, Hu Xu, Hugo Touvron, Iliyan Zarov, Imanol Arrieta Ibarra, Isabel Kloumann, Ishan Misra, Ivan Evtimov, Jack Zhang, Jade Copet, Jaewon Lee, Jan Geffert, Jana Vranes, Jason Park, Jay Mahadeokar, Jeet Shah, Jelmer van der Linde, Jennifer Billock, Jenny Hong, Jenya Lee, Jeremy Fu, Jianfeng Chi, Jianyu Huang, Jiawen Liu, Jie Wang, Jiecao Yu, Joanna Bitton, Joe Spisak, Jongsoo Park, Joseph Rocca, Joshua Johnstun, Joshua Saxe, Junteng Jia, Kalyan Vasuden Alwala, Karthik Prasad, Kartikeya Upasani, Kate Plawiak, Ke Li, Kenneth Heafield, Kevin Stone, Khalid El-Arini, Krithika Iyer, Kshitiz Malik, Kuenley Chiu, Kunal Bhalla, Kushal Lakhotia, Lauren Rantala-Yeary, Laurens van der Maaten, Lawrence Chen, Liang Tan, Liz Jenkins, Louis Martin, Lovish Madaan, Lubo Malo, Lukas Blecher, Lukas Landzaat, Luke de Oliveira, Madeline Muzzi, Mahesh Pasupuleti, Mannat Singh, Manohar Paluri, Marcin Kardas, Maria Tsimpoukelli, Mathew Oldham, Mathieu Rita, Maya Pavlova, Melanie Kambadur, Mike Lewis, Min Si, Mitesh Kumar Singh, Mona Hassan, Naman Goyal, Narjes Torabi, Nikolay Bashlykov, Nikolay Bogoychev, Niladri Chatterji, Ning Zhang, Olivier Duchenne, Onur Çelebi, Patrick Alrassy, Pengchuan Zhang, Pengwei Li, Petar Vasic, Peter Weng, Prajjwal Bhargava, Pratik Dubal, Praveen Krishnan, Punit Singh Koura, Puxin Xu, Qing He, Qingxiao Dong, Ragavan Srinivasan, Raj Ganapathy, Ramon Calderer, Ricardo Silveira Cabral, Robert Stojnic, Roberta Raileanu, Rohan Maheswari, Rohit Girdhar, Rohit Patel, Romain Sauvestre, Ronnie Polidoro, Roshan Sumbaly, Ross Taylor, Ruan Silva, Rui Hou, Rui

Wang, Saghar Hosseini, Sahana Chennabasappa, Sanjay Singh, Sean Bell, Seohyun Sonia Kim, Sergey Edunov, Shaoliang Nie, Sharan Narang, Sharath Raparthy, Sheng Shen, Shengye Wan, Shruti Bhosale, Shun Zhang, Simon Vandenhende, Soumya Batra, Spencer Whitman, Sten Sootla, Stephane Collot, Suchin Gururangan, Sydney Borodinsky, Tamar Herman, Tara Fowler, Tarek Sheasha, Thomas Georgiou, Thomas Scialom, Tobias Speckbacher, Todor Mihaylov, Tong Xiao, Ujjwal Karn, Vedanuj Goswami, Vibhor Gupta, Vignesh Ramanathan, Viktor Kerkez, Vincent Gonguet, Virginie Do, Vish Vogeti, Vítor Albiero, Vladan Petrovic, Weiwei Chu, Wenhan Xiong, Wenyin Fu, Whitney Meers, Xavier Martinet, Xiaodong Wang, Xiaofang Wang, Xiaoqing Ellen Tan, Xide Xia, Xinfeng Xie, Xuchao Jia, Xuewei Wang, Yaelle Goldschlag, Yashesh Gaur, Yasmine Babaei, Yi Wen, Yiwen Song, Yuchen Zhang, Yue Li, Yuning Mao, Zacharie Delpierre Coudert, Zheng Yan, Zhengxing Chen, Zoe Papakipos, Aaditya Singh, Aayushi Srivastava, Abha Jain, Adam Kelsey, Adam Shajnfeld, Adithya Gangidi, Adolfo Victoria, Ahuva Goldstand, Ajay Menon, Ajay Sharma, Alex Boesenberg, Alexei Baevski, Allie Feinstein, Amanda Kallet, Amit Sangani, Amos Teo, Anam Yunus, Andrei Lupu, Andres Alvarado, Andrew Caples, Andrew Gu, Andrew Ho, Andrew Poulton, Andrew Ryan, Ankit Ramchandani, Annie Dong, Annie Franco, Anuj Goyal, Aparajita Saraf, Arkabandhu Chowdhury, Ashley Gabriel, Ashwin Bharambe, Assaf Eisenman, Azadeh Yazdan, Beau James, Ben Maurer, Benjamin Leonhardi, Bernie Huang, Beth Loyd, Beto De Paola, Bhargavi Paranjape, Bing Liu, Bo Wu, Boyu Ni, Braden Hancock, Bram Wasti, Brandon Spence, Brani Stojkovic, Brian Gamido, Britt Montalvo, Carl Parker, Carly Burton, Catalina Mejia, Ce Liu, Changhan Wang, Changkyu Kim, Chao Zhou, Chester Hu, Ching-Hsiang Chu, Chris Cai, Chris Tindal, Christoph Feichtenhofer, Cynthia Gao, Damon Civin, Dana Beaty, Daniel Kreymer, Daniel Li, David Adkins, David Xu, Davide Testuggine, Delia David, Devi Parikh, Diana Liskovich, Didem Foss, Dingkang Wang, Duc Le, Dustin Holland, Edward Dowling, Eissa Jamil, Elaine Montgomery, Eleonora Presani, Emily Hahn, Emily Wood, Eric-Tuan Le, Erik Brinkman, Esteban Arcaute, Evan Dunbar, Evan Smothers, Fei Sun, Felix Kreuk, Feng Tian, Filippos Kokkinos, Firat Ozgenel, Francesco Caggioni, Frank Kanayet, Frank Seide, Gabriela Medina Florez, Gabriella Schwarz, Gada Badeer, Georgia Swee, Gil Halpern, Grant Herman, Grigory Sizov, Guangyi, Zhang, Guna Lakshminarayanan, Hakan Inan, Hamid Shojanazeri, Han Zou, Hannah Wang, Hanwen Zha, Haroun Habeeb, Harrison Rudolph, Helen Suk, Henry Aspegren, Hunter Goldman, Hongyuan Zhan, Ibrahim Damlaj, Igor Molybog, Igor Tufanov, Ilias Leontiadis, Irina-Elena Veliche, Itai Gat, Jake Weissman, James Geboski, James Kohli, Janice Lam, Japhet Asher, Jean-Baptiste Gaya, Jeff Marcus, Jeff Tang, Jennifer Chan, Jenny Zhen, Jeremy Reizenstein, Jeremy Teboul, Jessica Zhong, Jian Jin, Jingyi Yang, Joe Cummings, Jon Carvill, Jon Shepard, Jonathan McPhie, Jonathan Torres, Josh Ginsburg, Junjie Wang, Kai Wu, Kam Hou U, Karan Saxena, Kartikay Khandelwal, Katayoun Zand, Kathy Matosich, Kaushik Veeraraghavan, Kelly Michelena, Keqian Li, Kiran Jagadeesh, Kun Huang, Kunal Chawla, Kyle Huang, Lailin Chen, Lakshya Garg, Lavender A, Leandro Silva, Lee Bell, Lei Zhang, Liangpeng Guo, Licheng Yu, Liron Moshkovich, Luca Wehrstedt, Madian Khabsa, Manav Avalani, Manish Bhatt, Martynas Mankus, Matan Hasson, Matthew Lennie, Matthias Reso, Maxim Groshev, Maxim Naumov, Maya Lathi, Meghan Keneally, Miao Liu, Michael L. Seltzer, Michal Valko, Michelle Restrepo, Mihir Patel, Mik Vyatskov, Mikayel Samvelyan, Mike Clark, Mike Macey, Mike Wang, Miquel Jubert Hermoso, Mo Metanat, Mohammad Rastegari, Munish Bansal, Nandhini Santhanam, Natascha Parks, Natasha White, Navyata Bawa, Nayan Singhal, Nick Egebo, Nicolas Usunier, Nikhil Mehta, Nikolay Pavlovich Laptev, Ning Dong, Norman Cheng, Oleg Chernoguz, Olivia Hart, Omkar Salpekar, Ozlem Kalinli, Parkin Kent, Parth Parekh, Paul Saab, Pavan Balaji, Pedro Rittner, Philip Bontrager, Pierre Roux, Piotr Dollar, Polina Zvyagina, Prashant Ratanchandani, Pritish Yuvraj, Qian Liang, Rachad Alao, Rachel Rodriguez, Rafi Ayub, Raghotham Murthy, Raghu Nayani, Rahul Mitra, Rangaprabhu Parthasarathy, Raymond Li, Rebekkah Hogan, Robin Battey, Rocky Wang, Russ Howes, Ruty Rinott, Sachin Mehta, Sachin Siby, Sai Jayesh Bondu, Samyak Datta, Sara Chugh, Sara Hunt, Sargun Dhillon, Sasha Sidorov, Satadru Pan, Saurabh Mahajan, Saurabh Verma, Seiji Yamamoto, Sharadh Ramaswamy, Shaun Lindsay, Shaun Lindsay, Sheng Feng, Shenghao Lin, Shengxin Cindy Zha, Shishir Patil, Shiva Shankar, Shuqiang Zhang, Shuqiang Zhang, Sinong Wang, Sneha Agarwal, Soji Sajuyigbe, Soumith Chintala, Stephanie Max, Stephen Chen, Steve Kehoe, Steve Satterfield, Sudarshan Govin-

daprasad, Sumit Gupta, Summer Deng, Sungmin Cho, Sunny Virk, Suraj Subramanian, Sy Choudhury, Sydney Goldman, Tal Remez, Tamar Glaser, Tamara Best, Thilo Koehler, Thomas Robinson, Tianhe Li, Tianjun Zhang, Tim Matthews, Timothy Chou, Tzook Shaked, Varun Vontimitta, Victoria Ajayi, Victoria Montanez, Vijai Mohan, Vinay Satish Kumar, Vishal Mangla, Vlad Ionescu, Vlad Poenaru, Vlad Tiberiu Mihailescu, Vladimir Ivanov, Wei Li, Wenchen Wang, Wenwen Jiang, Wes Bouaziz, Will Constable, Xiaocheng Tang, Xiaojian Wu, Xiaolan Wang, Xilun Wu, Xinbo Gao, Yaniv Kleinman, Yanjun Chen, Ye Hu, Ye Jia, Ye Qi, Yenda Li, Yilin Zhang, Ying Zhang, Yossi Adi, Youngjin Nam, Yu, Wang, Yu Zhao, Yuchen Hao, Yundi Qian, Yunlu Li, Yuzi He, Zach Rait, Zachary DeVito, Zef Rosnbrick, Zhaoduo Wen, Zhenyu Yang, Zhiwei Zhao, and Zhiyu Ma. The llama 3 herd of models, 2024. URL https://arxiv.org/abs/2407.21783.

Daya Guo, Dejian Yang, Haowei Zhang, Junxiao Song, Peiyi Wang, Qihao Zhu, Runxin Xu, Ruoyu Zhang, Shirong Ma, Xiao Bi, Xiaokang Zhang, Xingkai Yu, Yu Wu, Z. F. Wu, Zhibin Gou, Zhihong Shao, Zhuoshu Li, Ziyi Gao, Aixin Liu, Bing Xue, Bingxuan Wang, Bochao Wu, Bei Feng, Chengda Lu, Chenggang Zhao, Chengqi Deng, Chong Ruan, Damai Dai, Deli Chen, Dongjie Ji, Erhang Li, Fangyun Lin, Fucong Dai, Fuli Luo, Guangbo Hao, Guanting Chen, Guowei Li, H. Zhang, Hanwei Xu, Honghui Ding, Huazuo Gao, Hui Qu, Hui Li, Jianzhong Guo, Jiashi Li, Jingchang Chen, Jingyang Yuan, Jinhao Tu, Junjie Qiu, Junlong Li, J. L. Cai, Jiaqi Ni, Jian Liang, Jin Chen, Kai Dong, Kai Hu, Kaichao You, Kaige Gao, Kang Guan, Kexin Huang, Kuai Yu, Lean Wang, Lecong Zhang, Liang Zhao, Litong Wang, Liyue Zhang, Lei Xu, Leyi Xia, Mingchuan Zhang, Minghua Zhang, Minghui Tang, Mingxu Zhou, Meng Li, Miaojun Wang, Mingming Li, Ning Tian, Panpan Huang, Peng Zhang, Qiancheng Wang, Qinyu Chen, Qiushi Du, Ruiqi Ge, Ruisong Zhang, Ruizhe Pan, Runji Wang, R. J. Chen, R. L. Jin, Ruyi Chen, Shanghao Lu, Shangyan Zhou, Shanhuang Chen, Shengfeng Ye, Shiyu Wang, Shuiping Yu, Shunfeng Zhou, Shuting Pan, S. S. Li, Shuang Zhou, Shaoqing Wu, Tao Yun, Tian Pei, Tianyu Sun, T. Wang, Wangding Zeng, Wen Liu, Wenfeng Liang, Wenjun Gao, Wenqin Yu, Wentao Zhang, W. L. Xiao, Wei An, Xiaodong Liu, Xiaohan Wang, Xiaokang Chen, Xiaotao Nie, Xin Cheng, Xin Liu, Xin Xie, Xingchao Liu, Xinyu Yang, Xinyuan Li, Xuecheng Su, Xuheng Lin, X. Q. Li, Xiangyue Jin, Xiaojin Shen, Xiaosha Chen, Xiaowen Sun, Xiaoxiang Wang, Xinnan Song, Xinyi Zhou, Xianzu Wang, Xinxia Shan, Y. K. Li, Y. Q. Wang, Y. X. Wei, Yang Zhang, Yanhong Xu, Yao Li, Yao Zhao, Yaofeng Sun, Yaohui Wang, Yi Yu, Yichao Zhang, Yifan Shi, Yiliang Xiong, Ying He, Yishi Piao, Yisong Wang, Yixuan Tan, Yiyang Ma, Yiyuan Liu, Yongqiang Guo, Yuan Ou, Yuduan Wang, Yue Gong, Yuheng Zou, Yujia He, Yunfan Xiong, Yuxiang Luo, Yuxiang You, Yuxuan Liu, Yuyang Zhou, Y. X. Zhu, Yanping Huang, Yaohui Li, Yi Zheng, Yuchen Zhu, Yunxian Ma, Ying Tang, Yukun Zha, Yuting Yan, Z. Z. Ren, Zehui Ren, Zhangli Sha, Zhe Fu, Zhean Xu, Zhenda Xie, Zhengyan Zhang, Zhewen Hao, Zhicheng Ma, Zhigang Yan, Zhiyu Wu, Zihui Gu, Zijia Zhu, Zijun Liu, Zilin Li, Ziwei Xie, Ziyang Song, Zizheng Pan, Zhen Huang, Zhipeng Xu, Zhongyu Zhang, and Zhen Zhang. Deepseek-r1 incentivizes reasoning in llms through reinforcement learning. *Nature*, 645(8081):633–638, September 2025. ISSN 1476-4687. doi: 10.1038/s41586-025-09422-z. URL http://dx.doi.org/10.1038/s41586-025-09422-z.

Xinyu He and Xianghui Cao. Understanding post-training structural changes in large language models, 2026. URL https://arxiv.org/abs/2509.17866.

Edward J Hu, yelong shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations*, 2022. URL https://openreview.net/forum?id=nZeVKeeFYf9.

Maggie Huan, Yuetai Li, Tuney Zheng, Xiaoyu Xu, Seungone Kim, Minxin Du, Radha Poovendran, Graham Neubig, and Xiang Yue. Does math reasoning improve general llm capabilities? understanding transferability of llm reasoning, 2025. URL https://arxiv.org/abs/2507.00432.

Gabriel Ilharco, Marco Tulio Ribeiro, Mitchell Wortsman, Ludwig Schmidt, Hannaneh Hajishirzi, and Ali Farhadi. Editing models with task arithmetic. In *The Eleventh International Conference on Learning Representations*, 2023. URL https://openreview.net/forum?id=6t0Kwf8-jrj.

Albert Q. Jiang, Alexandre Sablayrolles, Arthur Mensch, Chris Bamford, Devendra Singh Chaplot, Diego de las Casas, Florian Bressand, Gianna Lengyel, Guillaume Lample, Lucile Saulnier, Lélio Renard Lavaud, Marie-Anne Lachaux, Pierre Stock, Teven Le Scao, Thibaut Lavril, Thomas Wang, Timothée Lacroix, and William El Sayed. Mistral 7b, 2023. URL https://arxiv.org/abs/2310.06825.

Hangzhan Jin, Sitao Luan, Sicheng Lyu, Guillaume Rabusseau, Reihaneh Rabbany, Doina Precup, and Mohammad Hamdaqa. Rl fine-tuning heals ood forgetting in sft, 2025. URL https://arxiv.org/abs/2509.12235.

Jared Kaplan, Sam McCandlish, Tom Henighan, Tom B. Brown, Benjamin Chess, Rewon Child, Scott Gray, Alec Radford, Jeffrey Wu, and Dario Amodei. Scaling laws for neural language models, 2020. URL https://arxiv.org/abs/2001.08361.

Vijay Lingam, Atula Tejaswi Neerkaje, Aditya Vavre, Aneesh Shetty, Gautham Krishna Gudur, Joydeep Ghosh, Eunsol Choi, Alex Dimakis, Aleksandar Bojchevski, and sujay sanghavi. SVFT: Parameter-efficient fine-tuning with singular vectors. In *2nd Workshop on Advancing Neural Network Training: Computational Efficiency, Scalability, and Resource Optimization (WANT@ICML 2024)*, 2024. URL https://openreview.net/forum?id=DOUskwCqg5.

Mingjie Liu, Shizhe Diao, Ximing Lu, Jian Hu, Xin Dong, Yejin Choi, Jan Kautz, and Yi Dong. Prorl: Prolonged reinforcement learning expands reasoning boundaries in large language models, 2025. URL https://arxiv.org/abs/2505.24864.

Shih-Yang Liu, Chien-Yi Wang, Hongxu Yin, Pavlo Molchanov, Yu-Chiang Frank Wang, Kwang-Ting Cheng, and Min-Hung Chen. Dora: weight-decomposed low-rank adaptation. In *Proceedings of the 41st International Conference on Machine Learning*, ICML'24. JMLR.org, 2024.

Michael Luo, Sijun Tan, Justin Wong, Xiaoxiang Shi, William Y. Tang, Manan Roongta, Colin Cai, Jeffrey Luo, Li Erran Li, Raluca Ada Popa, and Ion Stoica. Deepscaler: Surpassing o1-preview with a 1.5b model by scaling rl. https://pretty-radio-b75.notion.site/DeepScaleR-Surpassing-O1-Preview-with-a-1-5B-Model-by-Scaling-RL-19681902c1468005bed8ca303013a4e2, 2025. Notion Blog.

Charles H. Martin and Michael W. Mahoney. Implicit self-regularization in deep neural networks: Evidence from random matrix theory and implications for learning, 2018. URL https://arxiv.org/abs/1810.01075.

Fanxu Meng, Zhaohui Wang, and Muhan Zhang. PiSSA: Principal singular values and singular vectors adaptation of large language models. In *The Thirty-eighth Annual Conference on Neural Information Processing Systems*, 2024. URL https://openreview.net/forum?id=6ZBHIEtdP4.

Sagnik Mukherjee, Lifan Yuan, Dilek Hakkani-Tür, and Hao Peng. Reinforcement learning finetunes small subnetworks in large language models. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems*, 2025a. URL https://openreview.net/forum?id=0NdS4xCngO.

Sagnik Mukherjee, Lifan Yuan, Dilek Hakkani-Tur, and Hao Peng. Reinforcement learning finetunes small subnetworks in large language models, 2025b. URL https://arxiv.org/abs/2505.11711.

Team Olmo, :, Allyson Ettinger, Amanda Bertsch, Bailey Kuehl, David Graham, David Heineman, Dirk Groeneveld, Faeze Brahman, Finbarr Timbers, Hamish Ivison, Jacob Morrison, Jake Poznanski, Kyle Lo, Luca Soldaini, Matt Jordan, Mayee Chen, Michael Noukhovitch, Nathan Lambert, Pete Walsh, Pradeep Dasigi, Robert Berry, Saumya Malik, Saurabh Shah, Scott Geng, Shane Arora, Shashank Gupta, Taira Anderson, Teng Xiao, Tyler Murray, Tyler Romero, Victoria Graf, Akari Asai, Akshita Bhagia, Alexander Wettig, Alisa Liu, Aman Rangapur, Chloe Anastasiades, Costa Huang, Dustin Schwenk, Harsh Trivedi, Ian Magnusson, Jaron Lochner, Jiacheng Liu, Lester James V. Miranda, Maarten

Sap, Malia Morgan, Michael Schmitz, Michal Guerquin, Michael Wilson, Regan Huff, Ronan Le Bras, Rui Xin, Rulin Shao, Sam Skjonsberg, Shannon Zejiang Shen, Shuyue Stella Li, Tucker Wilde, Valentina Pyatkin, Will Merrill, Yapei Chang, Yuling Gu, Zhiyuan Zeng, Ashish Sabharwal, Luke Zettlemoyer, Pang Wei Koh, Ali Farhadi, Noah A. Smith, and Hannaneh Hajishirzi. Olmo 3, 2025. URL https://arxiv.org/abs/2512.13961.

Team OLMo, Pete Walsh, Luca Soldaini, Dirk Groeneveld, Kyle Lo, Shane Arora, Akshita Bhagia, Yuling Gu, Shengyi Huang, Matt Jordan, Nathan Lambert, Dustin Schwenk, Oyvind Tafjord, Taira Anderson, David Atkinson, Faeze Brahman, Christopher Clark, Pradeep Dasigi, Nouha Dziri, Allyson Ettinger, Michal Guerquin, David Heineman, Hamish Ivison, Pang Wei Koh, Jiacheng Liu, Saumya Malik, William Merrill, Lester James V. Miranda, Jacob Morrison, Tyler Murray, Crystal Nam, Jake Poznanski, Valentina Pyatkin, Aman Rangapur, Michael Schmitz, Sam Skjonsberg, David Wadden, Christopher Wilhelm, Michael Wilson, Luke Zettlemoyer, Ali Farhadi, Noah A. Smith, and Hannaneh Hajishirzi. 2 olmo 2 furious, 2025. URL https://arxiv.org/abs/2501.00656.

Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Gray, John Schulman, Jacob Hilton, Fraser Kelton, Luke Miller, Maddie Simens, Amanda Askell, Peter Welinder, Paul Christiano, Jan Leike, and Ryan Lowe. Training language models to follow instructions with human feedback. In Alice H. Oh, Alekh Agarwal, Danielle Belgrave, and Kyunghyun Cho, editors, *Advances in Neural Information Processing Systems*, 2022. URL https://openreview.net/forum?id=TG8KACxEON.

Qwen, :, An Yang, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chengyuan Li, Dayiheng Liu, Fei Huang, Haoran Wei, Huan Lin, Jian Yang, Jianhong Tu, Jianwei Zhang, Jianxin Yang, Jiaxi Yang, Jingren Zhou, Junyang Lin, Kai Dang, Keming Lu, Keqin Bao, Kexin Yang, Le Yu, Mei Li, Mingfeng Xue, Pei Zhang, Qin Zhu, Rui Men, Runji Lin, Tianhao Li, Tianyi Tang, Tingyu Xia, Xingzhang Ren, Xuancheng Ren, Yang Fan, Yang Su, Yichang Zhang, Yu Wan, Yuqiong Liu, Zeyu Cui, Zhenru Zhang, and Zihan Qiu. Qwen2.5 technical report, 2025. URL https://arxiv.org/abs/2412.15115.

Evani Radiya-Dixit and Xin Wang. How fine can fine-tuning be? learning efficient language models, 2020. URL https://arxiv.org/abs/2004.14129.

Rafael Rafailov, Archit Sharma, Eric Mitchell, Stefano Ermon, Christopher D. Manning, and Chelsea Finn. Direct preference optimization: Your language model is secretly a reward model, 2024. URL https://arxiv.org/abs/2305.18290.

John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms, 2017. URL https://arxiv.org/abs/1707.06347.

Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, Y. K. Li, Y. Wu, and Daya Guo. Deepseekmath: Pushing the limits of mathematical reasoning in open language models, 2024. URL https://arxiv.org/abs/2402.03300.

Reece S Shuttleworth, Jacob Andreas, Antonio Torralba, and Pratyusha Sharma. LoRA vs full fine-tuning: An illusion of equivalence. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems*, 2025. URL https://openreview.net/forum?id=xp7B8rkh7L.

Chongjie Si, Xuankun Yang, Muqing Liu, Yadao Wang, Xiaokang Yang, Wenbo Su, Bo Zheng, and Wei Shen. Weight spectra induced efficient model adaptation, 2025. URL https://arxiv.org/abs/2505.23099.

Zhenpeng Su, Leiyu Pan, Xue Bai, Dening Liu, Guanting Dong, Jiaming Huang, Wenping Hu, Fuzheng Zhang, Kun Gai, and Guorui Zhou. Klear-reasoner: Advancing reasoning capability via gradient-preserving clipping policy optimization, 2026. URL https://arxiv.org/abs/2508.07629.

Eshaan Tanwar, Deepak Nathani, William Yang Wang, and Tanmoy Chakraborty. Understanding the effects of domain finetuning on llms, 2025. URL https://arxiv.org/abs/2510.09359.

Lewis Tunstall, Edward Beeching, Nathan Lambert, Nazneen Rajani, Kashif Rasul, Younes Belkada, Shengyi Huang, Leandro von Werra, Clémentine Fourrier, Nathan Habib, Nathan Sarrazin, Omar Sanseviero, Alexander M. Rush, and Thomas Wolf. Zephyr: Direct distillation of lm alignment, 2023. URL https://arxiv.org/abs/2310.16944.

Hanqing Wang, Yixia Li, Shuo Wang, Guanhua Chen, and Yun Chen. MiLoRA: Harnessing minor singular components for parameter-efficient LLM finetuning. In Luis Chiruzzo, Alan Ritter, and Lu Wang, editors, *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 4823–4836, Albuquerque, New Mexico, April 2025. Association for Computational Linguistics. ISBN 979-8-89176-189-6. doi: 10.18653/v1/2025.naacl-long.248. URL https://aclanthology.org/2025.naacl-long.248/.

Jason Wei, Yi Tay, Rishi Bommasani, Colin Raffel, Barret Zoph, Sebastian Borgeaud, Dani Yogatama, Maarten Bosma, Denny Zhou, Donald Metzler, Ed H. Chi, Tatsunori Hashimoto, Oriol Vinyals, Percy Liang, Jeff Dean, and William Fedus. Emergent abilities of large language models. *Transactions on Machine Learning Research*, 2022. ISSN 2835-8856. URL https://openreview.net/forum?id=yzkSU5zdwD. Survey Certification.

Xuansheng Wu, Wenlin Yao, Jianshu Chen, Xiaoman Pan, Xiaoyang Wang, Ninghao Liu, and Dong Yu. From language modeling to instruction following: Understanding the behavior shift in LLMs after instruction tuning. In Kevin Duh, Helena Gomez, and Steven Bethard, editors, *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 2341–2369, Mexico City, Mexico, June 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.naacl-long.130. URL https://aclanthology.org/2024.naacl-long.130/.

LLM-Core Xiaomi, :, Bingquan Xia, Bowen Shen, Cici, Dawei Zhu, Di Zhang, Gang Wang, Hailin Zhang, Huaqiu Liu, Jiebao Xiao, Jinhao Dong, Liang Zhao, Peidian Li, Peng Wang, Shihua Yu, Shimao Chen, Weikun Wang, Wenhan Ma, Xiangwei Deng, Yi Huang, Yifan Song, Zihan Jiang, Bowen Ye, Can Cai, Chenhong He, Dong Zhang, Duo Zhang, Guoan Wang, Hao Tian, Haochen Zhao, Heng Qu, Hongshen Xu, Jun Shi, Kainan Bao, Kai Fang, Kang Zhou, Kangyang Zhou, Lei Li, Menghang Zhu, Nuo Chen, Qiantong Wang, Shaohui Liu, Shicheng Li, Shuhao Gu, Shuhuai Ren, Shuo Liu, Sirui Deng, Weiji Zhuang, Weiwei Lv, Wenyu Yang, Xin Zhang, Xing Yong, Xing Zhang, Xingchen Song, Xinzhe Xu, Xu Wang, Yihan Yan, Yu Tu, Yuanyuan Tian, Yudong Wang, Yue Yu, Zhenru Lin, Zhichao Song, and Zihao Yue. Mimo: Unlocking the reasoning potential of language model – from pretraining to posttraining, 2025. URL https://arxiv.org/abs/2505.07608.

An Yang, Beichen Zhang, Binyuan Hui, Bofei Gao, Bowen Yu, Chengpeng Li, Dayiheng Liu, Jianhong Tu, Jingren Zhou, Junyang Lin, Keming Lu, Mingfeng Xue, Runji Lin, Tianyu Liu, Xingzhang Ren, and Zhenru Zhang. Qwen2.5-math technical report: Toward mathematical expert model via self-improvement, 2024. URL https://arxiv.org/abs/2409.12122.

An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, Chengen Huang, Chenxu Lv, Chujie Zheng, Dayiheng Liu, Fan Zhou, Fei Huang, Feng Hu, Hao Ge, Haoran Wei, Huan Lin, Jialong Tang, Jian Yang, Jianhong Tu, Jianwei Zhang, Jianxin Yang, Jiaxi Yang, Jing Zhou, Jingren Zhou, Junyang Lin, Kai Dang, Keqin Bao, Kexin Yang, Le Yu, Lianghao Deng, Mei Li, Mingfeng Xue, Mingze Li, Pei Zhang, Peng Wang, Qin Zhu, Rui Men, Ruize Gao, Shixuan Liu, Shuang Luo, Tianhao Li, Tianyi Tang, Wenbiao Yin, Xingzhang Ren, Xinyu Wang, Xinyu Zhang, Xuancheng Ren, Yang Fan, Yang Su, Yichang Zhang, Yinger Zhang, Yu Wan, Yuqiong Liu, Zekun Wang, Zeyu Cui, Zhenru Zhang, Zhipeng Zhou, and Zihan Qiu. Qwen3 technical report, 2025. URL https://arxiv.org/abs/2505.09388.

David Yunis, Kumar Kshitij Patel, Samuel Wheeler, Pedro Savarese, Gal Vardi, Karen Livescu, Michael Maire, and Matthew R. Walter. Approaching deep learning through the spectral dynamics of weights, 2024. URL https://arxiv.org/abs/2408.11804.

Hanning Zhang, Jiarui Yao, Chenlu Ye, Wei Xiong, and Tong Zhang. Online-dpo-r1: Unlocking effective reasoning without the ppo overhead. https://efficient-unicorn-451.notion.site/Online-DPO-R1-Unlocking-Effective-Reasoning-Without-the-PPO-Overhead-1908b9a70e7b80c3bc83f4cf04b2f175?pvs=4, 2025a. Notion Blog.

Hanning Zhang, Jiarui Yao, Chenlu Ye, Wei Xiong, and Tong Zhang. Online-dpo-r1: Unlocking effective reasoning without the ppo overhead. https://efficient-unicorn-451.notion.site/Online-DPO-R1-Unlocking-Effective-Reasoning-Without-the-PPO-Overhead-1908b9a70e7b80c3bc83f4cf04b2f175?pvs=4, 2025b. Notion Blog.

Chunting Zhou, Pengfei Liu, Puxin Xu, Srini Iyer, Jiao Sun, Yuning Mao, Xuezhe Ma, Avia Efrat, Ping Yu, LILI YU, Susan Zhang, Gargi Ghosh, Mike Lewis, Luke Zettlemoyer, and Omer Levy. LIMA: Less is more for alignment. In *Thirty-seventh Conference on Neural Information Processing Systems*, 2023. URL https://openreview.net/forum?id=KBMOKmX2he.

Hanqing Zhu, Zhenyu Zhang, Hanxian Huang, DiJia Su, Zechun Liu, Jiawei Zhao, Igor Fedorov, Hamed Pirsiavash, Zhizhou Sha, Jinwon Lee, David Z. Pan, Zhangyang Wang, Yuandong Tian, and Kai Sheng Tai. The path not taken: Rlvr provably learns off the principals, 2025. URL https://arxiv.org/abs/2511.08567.

A Disclosure

A.1 LLM Usage Disclosure

We used large language models as research assistants throughout the project. Specifically, LLMs were used to support code implementation, assist in refining hypotheses and experimental design, aid in literature discovery, and help improve writing clarity and organization. We used OpenAI Codex and Anthropic Claude Code to assist with research engineering tasks, including implementing and debugging experimental code, launching and monitoring experiments, analyzing intermediate outputs, and maintaining experiment records. We additionally used the `delta-research` framework[1] to organize iterative hypothesis testing and experiment execution.

These tools substantially accelerated the research process. The core research ideas, problem formulation, and overall methodology were initiated and guided by the authors. All generated code, experimental results, and analyses were carefully reviewed, validated, and iterated upon by the authors. The authors take full responsibility for the correctness, originality, and integrity of all content in this paper.

A.2 Reproducibility Disclosure

We include code, scripts, and instructions necessary to reproduce our results.

The codebase includes data processing pipelines, model loading, training and evaluation scripts, and analysis code for reported metrics and figures. In particular, we provide scripts to recompute all weight-space analyses, including SVD-based decompositions and alignment measures, directly from model checkpoints.

B Complete Transitions Table

Table 1 lists the model chains analyzed in this study. Each chain starts from a pretrained base model and applies one or more post-training stages. The current inventory contains 12 chains labeled C1 through C12. Table 2 gives the complete inventory of all 29 incremental transitions. Later appendices additionally analyze cumulative base to post transitions derived from the same chains, which is why those sections report larger transition counts.

Table 1: Chain inventory. C6 and C10 share the same pretrained base (Qwen2.5-Math-7B) but use different post-training pipelines. † indicates training directly from the pretrained base rather than from the preceding SFT checkpoint.

Chain	Base Model (HF identifier)	Size	Pipeline
C1	Qwen/Qwen2.5-Math-1.5B (Yang et al., 2024)	1.5B	SFT → GRPO (×2)
C2	Qwen/Qwen3-8B-Base (Yang et al., 2025)	8B	SFT → GPPO
C3	Qwen/Qwen3-14B (Yang et al., 2025)	14B	SFT → GRPO
C4	Qwen/Qwen3-8B-Base (Yang et al., 2025)	8B	GRPO (direct)
C5	mistralai/Mistral-7B-v0.1 (Jiang et al., 2023)	7B	SFT → DPO
C6	Qwen/Qwen2.5-Math-7B (Yang et al., 2024)	7B	SFT → DPO / PPO† / RAFT†
C7	allenai/OLMo-2-1124-7B (OLMo et al., 2025)	7B	SFT → DPO → RLVR
C8	meta-llama/Meta-Llama-3-8B (Grattafiori et al., 2024)	8B	SFT → PPO
C9	deepseek-ai/deepseek-math-7b-base (Shao et al., 2024)	7B	SFT → GRPO
C10	Qwen/Qwen2.5-Math-7B (Yang et al., 2024)	7B	SFT → PRIME / PRIME†
C11	XiaomiMiMo/MiMo-7B-Base (Xiaomi et al., 2025)	7B	SFT → RL / RL†
C12	allenai/OLMo-3-1025-7B (Olmo et al., 2025)	7B	SFT → DPO+OlmoRL

[1] https://github.com/user074/delta-research

Table 2: Complete inventory of incremental transitions. Each row is one training step producing the listed model. "From" gives the full name of the immediate predecessor model. † marks custom checkpoints not publicly released on HuggingFace. Domain indicates the training objective: **Math** = mathematical reasoning, **General** = general instruction following, **Reasoning** = broad reasoning / thinking, **Agent** = agentic tool use, **Metacog.** = metacognitive self-reflection.

Chain	Family	Trained Model	Size	From	Method	Domain
C1	Qwen 2.5	DeepSeek-R1-Distill-Qwen-1.5B (Guo et al., 2025)	1.5B	Qwen2.5-Math-1.5B	SFT	Reasoning
C1	Qwen 2.5	DeepScaleR-1.5B-Preview (Luo et al., 2025)	1.5B	DeepSeek-R1-Distill-Qwen-1.5B	GRPO	Math
C1	Qwen 2.5	Nemotron-Reasoning-Qwen-1.5B (Liu et al., 2025)	1.5B	DeepSeek-R1-Distill-Qwen-1.5B	GRPO	Math
C2	Qwen 3	Klear-Reasoner-8B-SFT (Su et al., 2026)	8B	Qwen3-8B-Base	SFT	Reasoning
C2	Qwen 3	Klear-Reasoner-8B (Su et al., 2026)	8B	Klear-Reasoner-8B-SFT	GPPO	Reasoning
C3	Qwen 3	UniReason-Qwen3-14B-SFT (Huan et al., 2025)	14B	Qwen3-14B	SFT	Reasoning
C3	Qwen 3	UniReason-Qwen3-14B-RL (Huan et al., 2025)	14B	Qwen3-14B	GRPO	Reasoning
C4	Qwen 3	SkyRL-Agent-WebResearch-8B (Cao et al., 2025)	8B	Qwen3-8B-Base	GRPO	Agent
C5	Mistral	mistral-7b-sft-beta (Tunstall et al., 2023)	7B	Mistral-7B-v0.1	SFT	General
C5	Mistral	zephyr-7b-beta (Tunstall et al., 2023)	7B	mistral-7b-sft-beta	DPO	General
C6	Qwen 2.5	Qwen2.5-7B-SFT (Zhang et al., 2025b)	7B	Qwen2.5-Math-7B	SFT	Math
C6	Qwen 2.5	Qwen2.5-7B-DPO (Zhang et al., 2025b)	7B	Qwen2.5-7B-SFT	DPO	Math
C6	Qwen 2.5	Qwen2.5-7B-PPO-Zero (Zhang et al., 2025b)	7B	Qwen2.5-Math-7B	PPO	Math
C6	Qwen 2.5	Qwen2.5-7B-RAFT-Zero (Zhang et al., 2025b)	7B	Qwen2.5-Math-7B	RAFT	Math
C7	OLMo 2	OLMo-2-1124-7B-SFT (OLMo et al., 2025)	7B	OLMo-2-1124-7B	SFT	General
C7	OLMo 2	OLMo-2-1124-7B-DPO (OLMo et al., 2025)	7B	OLMo-2-1124-7B-SFT	DPO	General
C7	OLMo 2	OLMo-2-1124-7B-Instruct (OLMo et al., 2025)	7B	OLMo-2-1124-7B-DPO	RLVR	General
C8	LLaMA 3	Llama-3-8b-sft-mixture (Dong et al., 2024)	8B	Meta-Llama-3-8B	SFT	General
C8	LLaMA 3	Llama-3-8b-rlhf-100k (Dong et al., 2024)	8B	Llama-3-8b-sft-mixture	PPO	General
C9	DeepSeek	deepseek-math-7b-instruct (Shao et al., 2024)	7B	deepseek-math-7b-base	SFT	Math
C9	DeepSeek	deepseek-math-7b-rl (Shao et al., 2024)	7B	deepseek-math-7b-instruct	GRPO	Math
C10	Qwen 2.5	Eurus-2-7B-SFT (Cui et al., 2025)	7B	Qwen2.5-Math-7B	SFT	Math
C10	Qwen 2.5	Eurus-2-7B-PRIME (Cui et al., 2025)	7B	Eurus-2-7B-SFT	PRIME	Math
C10	Qwen 2.5	Eurus-2-7B-PRIME-Zero (Cui et al., 2025)	7B	Qwen2.5-Math-7B	PRIME	Math
C11	MiMo	MiMo-7B-SFT (Xiaomi et al., 2025)	7B	MiMo-7B-Base	SFT	Reasoning
C11	MiMo	MiMo-7B-RL (Xiaomi et al., 2025)	7B	MiMo-7B-SFT	RL	Reasoning
C11	MiMo	MiMo-7B-RL-Zero (Xiaomi et al., 2025)	7B	MiMo-7B-Base	RL	Reasoning
C12	OLMo 3	OLMo-3-7B-Think-SFT (Olmo et al., 2025)	7B	OLMo-3-1025-7B	SFT	Reasoning
C12	OLMo 3	OLMo-3-7B-Think (Olmo et al., 2025)	7B	OLMo-3-7B-Think-SFT	DPO+OlmoRL	Reasoning

C Spectral Invariance of post-training

post-training leaves the singular value spectrum of weight matrices almost unchanged, even when the update is large. This appendix reports results for all 29 transitions in our inventory: 13 post-SFT follow-up transitions, 6 from-base RL transitions, and 11 base→SFT transitions.

C.1 Metrics

For each checkpoint pair $(W_{\text{pre}}, W_{\text{post}})$, we summarize the update size by

$$r_\Delta = \frac{\|W_{\text{post}} - W_{\text{pre}}\|_F}{\|W_{\text{pre}}\|_F}, \tag{8}$$

where the Frobenius norms aggregate all seven decomposed matrices across all layers in the transition. Table 3 is sorted by r_Δ within each block.

At the matrix level, we then compute singular values with `svdvals` and report two complementary measures of spectral change.

Singular Value (SV) Cosine. For a matrix $W \in \mathbb{R}^{m\times n}$, let $k = \min(m, n)$ and define its ordered singular-value vectors as

$$\sigma(W) = \big(\sigma_1(W), \sigma_2(W), \ldots, \sigma_k(W)\big), \qquad \sigma_1(W) \geq \sigma_2(W) \geq \cdots \geq \sigma_k(W) \geq 0. \tag{9}$$

The entries of $\sigma(W)$ are scalar singular values. We define it as singular value vectors so we can compare across the matrices of their singular value spectrum similarity. They should not be confused with the left and right singular vectors contained in U and V.

The cosine similarity between the pre and post-training singular value vectors:

$$\text{SV cosine} = \cos(\boldsymbol{\sigma}_{\text{pre}}, \boldsymbol{\sigma}_{\text{post}}) = \frac{\boldsymbol{\sigma}_{\text{pre}} \cdot \boldsymbol{\sigma}_{\text{post}}}{\|\boldsymbol{\sigma}_{\text{pre}}\| \, \|\boldsymbol{\sigma}_{\text{post}}\|}. \tag{10}$$

SV cosine $= 1.0$ means the spectral shape is perfectly preserved. Because singular values follow a steep power law, σ_1 dominates the dot product, so this metric is most sensitive to changes in the largest singular values.

Maximum Relative Perturbation. For one matrix, the maximum relative perturbation is

$$\delta_{\max} = \max_i \frac{|\sigma_i^{\text{post}} - \sigma_i^{\text{pre}}|}{\sigma_1^{\text{pre}}}. \tag{11}$$

This normalizes each singular value change by the spectral scale σ_1^{pre} and gives a scale invariant worst case measure. We use SV cosine as a global summary and $\delta_{\max}$ as the stricter componentwise check.

Aggregation. Each transition contains hundreds of weight matrices (7 module types across all layers). For each transition we report the mean and standard deviation of both SV cosine and $\delta_{\max}$ across matrices.

C.2 Results

Table 3 reports results for all 29 transitions, with each block sorted by r_Δ.

Post-SFT follow-up transitions. All 12 post-SFT follow-up transitions achieve SV cosine $= 1.000000$ to 6 decimal places, with $\delta_{\max} < 0.04\%$. These updates are effectively isospectral. They redistribute weight energy across directions without meaningfully rescaling singular values.

Cumulative SFT. SV cosine is at least 0.9999 across all 11 transitions, including the most aggressive one, C2, where $r_\Delta = 0.29$ and SV cosine is still 0.99993. The $\delta_{\max}$ metric shows more variation with training intensity. Lighter SFT runs with $r_\Delta < 0.04$ perturb singular values by less than 0.3% of σ_1, while the most aggressive SFT reaches 8.6% in C2. Even there, no individual singular value moves by more than 8.6% of the matrix's spectral scale.

From-base RL. The four direct math-RL branches, PPO-Zero, RAFT-Zero, PRIME-Zero, and RL-Zero, also show SV cosine $= 1.000000$ and $\delta_{\max} < 0.04\%$. RL alone is therefore isospectral in these runs as well. The only larger from-base RL case is C4 direct GRPO: SV cosine is still 0.999999, but $\delta_{\max}$ rises to 1.29%.

C.3 Discussion

post-training Is Almost Isospectral Both SFT and RL preserve the singular value spectrum with high precision, even when they produce functionally different models. The clearest case is aggressive SFT: C2 base→SFT has $r_\Delta = 0.29$ yet SV cosine $= 0.99993$. The weight matrix moves a lot, but the global spectral shape hardly changes. post-training mainly redistributes energy across directions. It changes which features are amplified and adds out-of-frame components through the null term. It does not substantially rescale existing singular values. This matches the main text, where the diagonal term is the least important component in the causal tests.

D Effective Rank Is Invariant Under post-training

The null component $\Delta\mathbf{W}_{\text{null}}$ lies outside the pretrained singular vector frame $\mathcal{S} = \{\mathbf{U}_0 A \mathbf{V}_0^\top : A \in \mathbb{R}^{r \times r}\}$ and can therefore let a layer read from directions outside $\text{col}(\mathbf{V}_0)$

Table 3: Spectral invariance of weight matrices across 29 transitions: incremental post-SFT updates, from-base RL, and cumulative SFT, sorted by r_Δ within each group. SV cosine and $\delta_{\max}$ report mean $\pm$ std across all weight matrices in the transition.

Chain	Transition	r_Δ	SV cos	$\delta_{\max}$
Post-SFT follow-up transitions (12 transitions)				
C6 (Qwen2.5-7B)	SFT→DPO	0.0003	$1.000000_{\pm 0}$	$0.01_{\pm 0.01}\%$
C8 (LLaMA-3-8B)	SFT→PPO	0.0010	$1.000000_{\pm 0}$	$0.01_{\pm 0.01}\%$
C1 (Qwen2.5-1.5B)	SFT→GRPO-A	0.0010	$1.000000_{\pm 0}$	$0.01_{\pm 0.01}\%$
C10 (Eurus-2-7B)	SFT→PRIME	0.0010	$1.000000_{\pm 0}$	$0.02_{\pm 0.01}\%$
C2 (Qwen3-8B)	SFT→GPPO	0.0020	$1.000000_{\pm 0}$	$0.02_{\pm 0.01}\%$
C9 (DSMath-7B)	SFT→RL	0.0020	$1.000000_{\pm 0}$	$0.02_{\pm 0.01}\%$
C12 (OLMo-3-7B)	SFT→RL	0.0020	$1.000000_{\pm 0}$	$0.02_{\pm 0.01}\%$
C7 (OLMo-2-7B)	SFT→DPO	0.0020	$1.000000_{\pm 0}$	$0.02_{\pm 0.04}\%$
C7 (OLMo-2-7B)	SFT→Instruct	0.0020	$1.000000_{\pm 0}$	$0.03_{\pm 0.04}\%$
C11 (MiMo-7B)	SFT→RL	0.0030	$1.000000_{\pm 0}$	$0.03_{\pm 0.01}\%$
C1 (Qwen2.5-1.5B)	SFT→GRPO-B	0.0050	$1.000000_{\pm 0}$	$0.02_{\pm 0.02}\%$
C5 (Mistral-7B)	SFT→DPO	0.0070	$1.000000_{\pm 0}$	$0.03_{\pm 0.02}\%$
From-base RL: base → RL (5 transitions)				
C6 (Qwen2.5-7B)	base→PPO	0.0010	$1.000000_{\pm 0}$	$0.01_{\pm 0.01}\%$
C10 (Qwen2.5-7B)	base→PRIME-Zero	0.0010	$1.000000_{\pm 0}$	$0.01_{\pm 0.01}\%$
C11 (MiMo-7B)	base→RL-Zero	0.0020	$1.000000_{\pm 0}$	$0.02_{\pm 0.01}\%$
C6 (Qwen2.5-7B)	base→RAFT	0.0050	$1.000000_{\pm 0}$	$0.02_{\pm 0.01}\%$
C4 (Qwen3-8B)	base→GRPO	0.0810	$0.999999_{\pm 0.000002}$	$1.29_{\pm 0.29}\%$
C3 (Qwen3-14B)	base→GRPO	0.0825	$0.999999_{\pm 0.000002}$	$1.29_{\pm 0.32}\%$
Cumulative SFT: base → SFT (11 transitions)				
C6 (Qwen2.5-7B)	base→SFT	0.006	$1.000000_{\pm 0}$	$0.04_{\pm 0.04}\%$
C8 (LLaMA-3-8B)	base→SFT	0.019	$1.000000_{\pm 0}$	$0.08_{\pm 0.07}\%$
C10 (Eurus-2-7B)	base→SFT	0.024	$1.000000_{\pm 0}$	$0.11_{\pm 0.12}\%$
C9 (DSMath-7B)	base→SFT	0.030	$1.000000_{\pm 0}$	$0.24_{\pm 0.08}\%$
C5 (Mistral-7B)	base→SFT	0.038	$1.000000_{\pm 0.000001}$	$0.25_{\pm 0.23}\%$
C7 (OLMo-2-7B)	base→SFT	0.073	$0.999998_{\pm 0.000004}$	$0.62_{\pm 0.60}\%$
C3 (Qwen3-14B)	base→SFT	0.084	$0.999999_{\pm 0.000002}$	$1.29_{\pm 0.31}\%$
C11 (MiMo-7B)	base→SFT	0.189	$0.999953_{\pm 0.000075}$	$1.60_{\pm 0.75}\%$
C1 (Qwen2.5-1.5B)	base→SFT	0.192	$0.999979_{\pm 0.000037}$	$7.92_{\pm 11.45}\%$
C12 (OLMo-3-7B)	base→SFT	0.248	$0.999976_{\pm 0.000033}$	$5.05_{\pm 2.99}\%$
C2 (Qwen3-8B)	base→SFT	0.290	$0.999930_{\pm 0.000114}$	$8.63_{\pm 2.52}\%$

or write into directions outside $\mathrm{col}(\mathbf{U}_0)$. This raises a natural question: does post-training increase the effective dimensionality of the weight matrix?

Algebraic rank is not informative here. At floating point precision, these matrices are already numerically full rank, and tiny perturbations in the spectral tail can move many near-zero singular values across a machine-epsilon threshold without reflecting a meaningful structural change. We therefore focus on *effective rank*, namely how much of the spectral mass is concentrated in the dominant modes. In our data the stable rank of pretrained weights is much smaller than algebraic rank, typically between 109 and 184, so it can in principle change if post-training creates genuinely new dominant directions.

Metrics. We compute six rank metrics for $\mathbf{W}_0$, $\mathbf{W}_1 = \mathbf{W}_0 + \Delta\mathbf{W}$, and $\Delta\mathbf{W}$ across all matrices from 42 transition instances over 12 chains and 8 architecture families (1.5B–14B parameters). These are stable rank, ε-rank at $\varepsilon \in \{0.01, 0.001\}$, entropy erank, and the energy capture ranks k_{50} and k_{90}. We also define the **rank additivity ratio**

$$\alpha = (\mathrm{SR}(\mathbf{W}_1) - \mathrm{SR}(\mathbf{W}_0)) \,/\, \mathrm{SR}(\Delta\mathbf{W}),$$

which equals 1 if the update's rank adds to the weight's rank and 0 if it is absorbed.

Per-transition detail. Table 4 reports stable rank and additivity for all 42 transition instances. Two points stand out. First, rank is not additive. $\mathrm{SR}(\Delta\mathbf{W})$ ranges from 6.6

to 382, yet the median additivity ratio is $\alpha = -0.00002$. Even C11 MiMo SFT, where $\mathrm{SR}(\Delta\mathbf{W}) = 382$, transfers only 1.3% of the update's rank. The update's spectral energy is absorbed into the existing modes of $\mathbf{W}_0$, not stacked on top. Second, cumulative RL mirrors SFT. For example, C1 base→GRPO-A ($\Delta\mathrm{SR} = -0.60$) tracks C1 base→SFT ($-0.60$), so the incremental RL step adds almost no extra rank change. The same pattern holds across the larger chains: base→GPPO in C2, base→RL in C3, and base→RL in C12 all stay close to their corresponding base→SFT rank changes.

Table 4: Stable rank and additivity for all 42 transitions. $\mathrm{SR}_0/\mathrm{SR}_\Delta/\mathrm{SR}_1$: stable rank of $\mathbf{W}_0/\Delta\mathbf{W}/\mathbf{W}_1$; α: additivity ratio.

Transition	SR_0	SR_Δ	SR_1	$\Delta\%$	α
SFT (base→SFT)					
C1	109.8	82.3	109.2	−0.60	−0.007
C2	167.7	277.5	165.2	−1.49	+0.017
C3	183.1	197.2	183.5	+0.17	−0.002
C5	181.0	19.7	181.2	+0.07	+0.001
C6	176.6	10.7	176.3	−0.15	−0.001
C7	140.7	156.4	140.6	−0.09	−0.001
C8	154.4	63.7	154.5	+0.04	+0.000
C9	178.7	64.4	178.8	+0.05	−0.000
C10	176.6	59.5	176.7	+0.07	+0.000
C11	178.5	382.1	184.3	+3.21	+0.013
C12	133.9	200.4	130.7	−2.36	−0.007
Incremental RL (SFT→RL)					
C1: GRPO-A	109.2	11.3	109.2	+0.00	−0.000
C1: GRPO-B	109.2	18.6	109.2	+0.02	+0.000
C2: GPPO	165.2	89.4	165.2	+0.00	−0.000
C5: DPO	181.2	27.4	181.2	+0.02	−0.000
C6: DPO	176.3	24.0	176.3	−0.00	−0.000
C7: DPO	140.6	20.4	140.5	−0.05	+0.000
C7: Inst.	140.6	21.7	140.5	−0.01	+0.000
C8: PPO	154.5	19.6	154.4	−0.03	+0.000
C9: RL	178.8	18.3	178.8	+0.00	−0.000
C10: PRIME	176.7	10.8	176.7	+0.00	+0.000
C11: RL	184.3	69.3	184.2	−0.03	−0.000
C12: RL	130.7	25.7	130.7	−0.02	+0.000
Cumulative RL (base→RL)					
C1: GRPO-A	109.8	82.3	109.2	−0.60	−0.007
C1: GRPO-B	109.8	82.3	109.2	−0.58	−0.007
C2: GPPO	167.7	277.5	165.2	−1.48	+0.017
C3: RL	183.1	202.1	183.5	+0.18	−0.002
C4: direct GRPO	167.7	192.6	167.6	−0.07	+0.002
C5: DPO	181.0	20.0	181.2	+0.09	+0.001
C6: DPO	176.6	10.7	176.3	−0.15	−0.001
C6: PPO	176.6	29.8	176.6	+0.00	−0.000
C6: RAFT	176.6	20.7	176.6	−0.01	+0.000
C7: DPO	140.7	156.9	140.5	−0.13	−0.001
C7: Inst.	140.7	157.0	140.5	−0.10	−0.001
C8: PPO	154.4	63.5	154.4	+0.00	+0.000
C9: RL	178.7	64.5	178.8	+0.05	−0.000
C10: PRIME	176.6	59.5	176.7	+0.08	+0.000
C10: PRIME-Zero	176.6	6.6	176.6	+0.01	+0.000
C11: RL	178.5	382.1	184.2	+3.18	+0.013
C11: RL-Zero	178.5	28.1	178.5	−0.02	−0.000
C12: RL	133.9	200.4	130.7	−2.38	−0.007

A note on numerical rank. At machine epsilon ($\sim 1.2 \times 10^{-7}$), apparent rank changes of up to +1,439 occur (C7 OLMo-2 `k_proj`, layers 8–9) because ∼1,400 very small singular

values cross the threshold after training. At any $\varepsilon \geq 0.001$, the change drops to ≤ 87. Numerical rank at machine epsilon is not useful for this analysis.

D.1 Interpretation

The diagonal and off-diagonal components act within $\mathcal{S}$ and cannot change effective rank by construction. The null component accesses directions outside $\mathcal{S}$, but the rank invariance result shows that this does not create new dominant modes. Out-of-frame components introduced by $\Delta\mathbf{W}_{\text{null}}$ are absorbed into the spectral structure of $\mathbf{W}_0$ rather than expanding it. This is still a local statement about one layer. Residual paths mean a direction outside one layer's pretrained frame may already be represented elsewhere in the network. The narrower point is that $\Delta\mathbf{W}_{\text{null}}$ can access directions outside a given layer's pretrained frame without increasing that layer's effective dimensionality.

The broader lesson is simple. post-training can change *which* directions a layer reads from and writes to without increasing *how many* dominant modes it maintains. Together with the main text's finding that removing the diagonal usually does not hurt, this suggests that post-training is mainly a routing problem inside a spectral envelope fixed by pretraining.

E Causal Decomposition: Full Evaluation Results

This appendix gives the full causal decomposition results summarized in Section 4.2. For each transition, we reconstruct $\mathbf{W}_1 = \mathbf{W}_0 + \Delta\mathbf{W}$ using either a single component or the OffNull model that keeps the off and null pair from the P-matrix decomposition, then evaluate average accuracy across five math benchmarks: MATH500, AIME24, AMC23, Minerva Math, and OlympiadBench. "Retention" is accuracy relative to the full model.

Reading the tables. Each row is one post-training transition. **Full** is the unmodified $\mathbf{W}_1$. **Diag** keeps only $\Delta\mathbf{W}_{\text{diag}}$. **Off** keeps only the off-diagonal component. **Null** keeps only the null space component. **OffNull** removes the diagonal and keeps both routing terms. Values near 100% in the OffNull column mean the diagonal is not needed for downstream performance. Tables 5 and 6 separate SFT and RL transitions for readability.

Table 5: Causal decomposition: SFT transitions (base → SFT). Accuracy (%) averaged over five math benchmarks; retention relative to Full in parentheses.

Chain	Base → SFT model	Base	Full	Diag	Off	Null	OffNull
C1	Qwen2.5-1.5B → DS-R1-Distill	31.3	49.2	6.4 (13%)	31.3 (64%)	18.5 (37%)	28.9 (59%)
C2	Qwen3-8B → KlearReasoner-SFT	21.2	60.4	14.0 (23%)	9.5 (16%)	9.7 (16%)	56.1 (93%)
C3	Qwen3-14B → UniReason-SFT	17.2	49.1	28.0 (57%)	54.3 (111%)	55.7 (113%)	45.8 (93%)
C5	Mistral-7B → mistral-sft-beta	3.6	5.7	1.9 (33%)	5.8 (102%)	6.2 (110%)	5.7 (100%)
C6	Qwen2.5-Math-7B → RLHFlow-SFT	35.8	43.5	38.8 (89%)	49.7 (114%)	48.0 (110%)	42.1 (97%)
C7	OLMo-2-7B → SFT	5.0	8.6	7.9 (92%)	9.0 (105%)	9.8 (114%)	7.5 (88%)
C8	LLaMA-3-8B → OpenRLHF-SFT	4.5	10.0	2.2 (22%)	9.7 (97%)	7.6 (75%)	10.9 (109%)
C9	DSMath-7B → Instruct	4.6	22.1	5.6 (25%)	19.6 (89%)	17.8 (81%)	21.0 (95%)
C10	Qwen2.5-Math-7B → Eurus2-SFT	35.8	35.0	41.9 (120%)	39.9 (114%)	38.8 (111%)	34.0 (97%)
C11	MiMo-7B → MiMo-SFT	9.6	65.6	0.0 (0%)	57.6 (88%)	44.7 (68%)	66.8 (102%)
C12	OLMo-3-7B → Think-SFT	7.3	61.1	2.8 (5%)	40.9 (67%)	19.3 (32%)	62.0 (102%)

(*n*%) denotes retention %. OffNull $\geq 88\%$ for 10/11 chains: removing the diagonal preserves most or all performance. C1 (59%) is the sole exception, where the SFT is heavily distilled from DeepSeek-R1.

Summary statistics. Across all 28 transitions, OffNull retention has median 99% and mean 97%. On the 25 positive gain transitions used for the main text summaries, the OffNull model stays within 4.3 points of the full model in 24 cases. Across the full set, it is at least 95% in 22 of 28 transitions and at least 92% in 25 of 28. Diag only retention is much lower, with median 79% for RL and 23% for SFT. In 6 of 11 SFT chains, the diagonal alone retains at most 25% of the full performance. Null only retention is generally

Table 6: Causal decomposition: RL transitions (SFT → RL or base → RL). Same format as Table 5.

Chain	RL model	Alg.	Base	Full	Diag	Off	Null	OffNull
C1	DeepScaleR	GRPO	49.2	51.8	41.0 (79%)	43.8 (85%)	51.7 (100%)	54.4 (105%)
C1	Nemotron	GRPO	49.2	61.3	40.3 (66%)	57.6 (94%)	59.2 (97%)	62.7 (102%)
C2	KlearReasoner	GPPO	60.4	71.7	58.6 (82%)	69.8 (97%)	69.0 (96%)	72.6 (101%)
C3	UniReason	GRPO	49.1	66.5	75.1 (113%)	64.2 (96%)	65.9 (99%)	65.0 (98%)
C5	zephyr-beta	DPO	5.7	3.1	5.2 (168%)	5.3 (171%)	4.7 (153%)	3.7 (120%)
C6	DPO	DPO	43.5	49.7	41.2 (83%)	45.0 (90%)	48.5 (98%)	49.8 (100%)
C6	PPO-Zero	PPO	35.8	50.1	37.4 (75%)	43.6 (87%)	44.4 (89%)	49.7 (99%)
C6	RAFT-Zero	RAFT	35.8	44.6	38.4 (86%)	43.1 (97%)	44.3 (99%)	45.4 (102%)
C7	DPO	DPO	8.6	11.3	8.8 (79%)	9.3 (83%)	10.5 (94%)	11.7 (104%)
C7	RLVR	RLVR	11.3	11.7	11.9 (101%)	10.4 (89%)	12.2 (104%)	10.8 (93%)
C8	PPO	PPO	10.0	11.1	10.8 (97%)	10.1 (91%)	12.1 (109%)	11.2 (101%)
C9	RL	GRPO	22.1	25.5	19.2 (75%)	22.4 (88%)	21.2 (83%)	23.8 (93%)
C10	PRIME	PRIME	35.0	44.2	36.1 (82%)	39.7 (90%)	40.7 (92%)	44.3 (100%)
C10	PRIME-Zero	PRIME	35.8	42.1	35.0 (83%)	36.4 (86%)	41.2 (98%)	38.6 (92%)
C11	MiMo-RL	RL	65.6	69.1	67.3 (97%)	70.2 (102%)	69.7 (101%)	67.0 (97%)
C11	MiMo-RL-Zero	RL	9.6	49.7	10.5 (21%)	36.8 (74%)	39.9 (80%)	47.3 (95%)
C12	Think-RL	RL	61.1	68.2	61.2 (90%)	65.5 (96%)	64.9 (95%)	67.8 (99%)

OffNull $\geq$ 92% for 15/17 RL transitions. C5 zephyr-beta (120%) and C7 RLVR (93%) are the only cases below 95%; both involve very small relative updates where all components are near noise level.

higher than diagonal only retention for both SFT and RL, which matches the routing interpretation: null space changes contribute more to behavioral change than simple spectral rescaling.

F Energy Fraction Decomposition: Full Results

The main text shows that the diagonal component usually carries only a small fraction of the update energy, while the off-diagonal and null space terms carry most of it. This appendix gives the full transition by transition breakdown and explains where that split comes from.

Definition. For each transition we compute the energy fraction carried by each component: $f_{\text{diag}} = \|\Delta\mathbf{W}_{\text{diag}}\|_F^2 / \|\Delta\mathbf{W}\|_F^2$, and analogously f_{off} and f_{null}. By the Pythagorean property of the decomposition, the three fractions sum to 1.

Geometric constraint on the partition. For square projections ($m = n = r$), both $\text{col}(\mathbf{U}_0)$ and $\text{col}(\mathbf{V}_0)$ span their ambient spaces, so $\mathcal{S} = \mathbb{R}^{m\times n}$ and $\Delta\mathbf{W}_{\text{null}}$ vanishes identically. In that case the decomposition reduces to the in frame part studied by prior spectral analyses of post-training (Zhu et al., 2025; He and Cao, 2026). In practice, however, five of the seven projections per transformer layer are rectangular (gate, up, down, and k/v under GQA), and the null component typically carries 26–56% of the update energy.

Full results. Table 7 reports the mean energy fraction across all layers and module types for every transition.

Observations. Three patterns are clear across the 28 transitions. First, the diagonal is negligible for RL. All 17 RL transitions have $f_{\text{diag}} \leq 0.011$, and 14 of 17 have $f_{\text{diag}} \leq 0.001$. RL places almost all of its update energy in off-diagonal rotation and null space redirection. Second, SFT shows a wider range. Two SFT chains, C1 and C2, allocate 15–20% of their energy to the diagonal. These are the only transitions with substantial f_{diag}, and both involve aggressive distillation with large relative updates. The other 9 SFT chains have

Table 7: Mean energy fraction of each $\Delta\mathbf{W}$ component, averaged over all layers and module types. Fractions sum to 1 by construction.

Chain	Transition	f_{diag}	f_{off}	f_{null}
SFT (base→SFT)				
C1	Qwen2.5-1.5B → DS-R1-Distill	0.204	0.288	0.508
C2	Qwen3-8B → KlearReasoner-SFT	0.149	0.456	0.397
C3	Qwen3-14B → UniReason-SFT	0.011	0.504	0.487
C5	Mistral-7B → mistral-sft-beta	0.000	0.508	0.492
C6	Qwen2.5-Math-7B → RLHFlow-SFT	0.000	0.453	0.548
C7	OLMo-2-7B → SFT	0.000	0.736	0.265
C8	LLaMA-3-8B → OpenRLHF-SFT	0.000	0.499	0.501
C9	DSMath-7B → Instruct	0.007	0.734	0.260
C10	Qwen2.5-Math-7B → Eurus2-SFT	0.000	0.444	0.556
C11	MiMo-7B → MiMo-SFT	0.005	0.525	0.471
C12	OLMo-3-7B → Think-SFT	0.003	0.735	0.263
RL (both incremental SFT→RL and base→RL)				
C1	→ DeepScaleR (GRPO)	0.001	0.450	0.549
C1	→ Nemotron (GRPO)	0.001	0.438	0.562
C2	→ KlearReasoner (GPPO)	0.001	0.531	0.469
C3	→ UniReason (GRPO)	0.011	0.504	0.487
C4	→ SkyRL-Agent (direct GRPO)	0.033	0.518	0.451
C5	→ zephyr-beta (DPO)	0.000	0.507	0.494
C6	→ DPO	0.000	0.416	0.585
C6	→ PPO-Zero	0.000	0.427	0.573
C6	→ RAFT-Zero	0.000	0.441	0.559
C7	→ DPO	0.001	0.737	0.264
C7	→ RLVR	0.000	0.734	0.268
C8	→ PPO	0.000	0.488	0.513
C9	→ RL (GRPO)	0.000	0.739	0.263
C10	→ PRIME	0.000	0.460	0.541
C10	→ PRIME-Zero	0.000	0.456	0.545
C11	→ MiMo-RL	0.005	0.539	0.457
C11	→ MiMo-RL-Zero	0.002	0.533	0.466
C12	→ Think-RL	0.000	0.738	0.263

$f_{\text{diag}} \leq 0.011$, which is indistinguishable from RL. Third, the split between off-diagonal and null space is governed mostly by matrix shape, not by algorithm. Chains C7, C9, and C12 consistently show $f_{\text{off}} \approx 0.73$ and $f_{\text{null}} \approx 0.26$ whether the transition is SFT or RL. By contrast, chains with more rectangular projections, such as C1, C6, and C10, show $f_{\text{null}} > 0.5$.

G Cross-Method Comparisons from a Shared Base

The sequential analysis above compared SFT and RL inside one pipeline. Here we ask a different question: when different groups post-train the *same* base model with different algorithms, data, or recipes, how similar are the resulting updates?

This comparison is possible because GRRR projects all updates into the same pretrained SVD basis. We study eight pairs across three settings: RL against RL, SFT against RL, and SFT against SFT from a shared base. For each pair, we report the Frobenius norm of each update and the off-diagonal cosine between their P-matrices. Results appear in Figure 5.

Independent methods show low but structured alignment. Excluding C3, the off-diagonal cosine ranges from 0.003 to 0.146. Figure 5B shows the two-sigma random baseline in gray. Every pair lies above that envelope. The observed values are small in absolute terms but they are not random. This suggests that independent post-training methods share a weak but real directional bias from the base model and task domain. The pattern appears in all three comparison types: different RL algorithms from the same checkpoint,

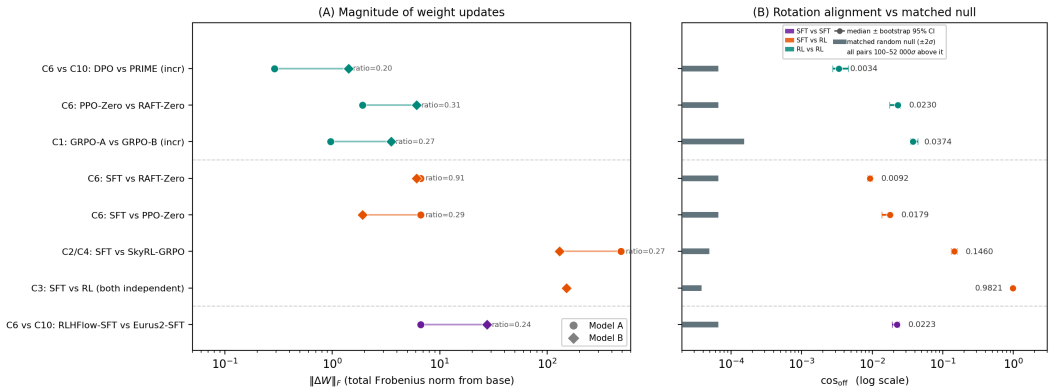


Figure 5: **Cross method comparison: same base, different training. (A)** Frobenius norms of the two updates (log scale), with the smaller to larger ratio annotated. **(B)** Off-diagonal cosine similarity. The matched isotropic null accounts for each pair's matrix shapes and the aggregation statistic used in the figure. All pairs are positively aligned beyond this null. The sole exception is C3, where independently trained SFT and RL produce nearly identical rotations ($\cos = 0.98$) and closely matched update magnitudes.

SFT compared against independent RL from the same base, and different SFT recipes from the same base.

Comparison type itself does not predict alignment very well. RL against RL pairs range from 0.003 to 0.037, SFT against RL pairs range from 0.009 to 0.146, and the single SFT against SFT pair is 0.022. The variation within each category is comparable to the variation between categories. This suggests that data and training scale matter more than the SFT/RL label when determining update direction.

C3: controlled data and compute produce identical updates. The striking exception is C3 (Qwen3-14B), where UniReason-SFT and UniReason-RL are trained independently from the same base with controlled data and computation. Their off-diagonal cosine is 0.982, and Figure 5A shows that their update norms are also closely matched. The updates are nearly identical in both direction and magnitude even though one uses SFT and the other uses GRPO. This shows that when training data and optimization budget are matched, the algorithm label has almost no effect on update direction. The base model and data determine *where* post-training moves in weight space. The algorithm mostly determines *how* it gets there.

Comparison with the sequential pipeline. These cross method results complement the sequential analysis. In sequential pipelines, where RL starts from the SFT checkpoint, the off-diagonal cosine stays below 0.008. In shared base comparisons, the cosine is higher, between 0.003 and 0.146. The reason is geometric. Sequential orthogonality appears because RL is initialized from SFT and therefore moves into directions that SFT left unused. Cross method alignment is low for a different reason: independent training trajectories diverge in high dimensions even when they share some directional bias from the same base.

The C3 result ties these observations together. When data and compute are controlled, even the SFT/RL distinction largely disappears. The near orthogonality of the sequential pipeline should therefore not be read as an intrinsic algorithmic difference. It mainly reflects the fact that RL starts from the SFT checkpoint and is constrained to a different part of the loss landscape.

H Random subspace study

Transition	Full	Remove diag.	Remove random	Δ_{diag}	Δ_{rand}
C1 SFT	49.2	28.9	44.2	-20.3	-5.0
C2 SFT	60.4	56.1	45.2	-4.3	-15.2

Table 8: Removing the learned diagonal component compared with removing a random component of the same Frobenius norm within the pretrained SVD core. Random-removal results are averaged over three seeds.